



UNDERSTANDING GLOBAL FOOD WASTE

A Global Analysis of Food Waste Patterns,
Drivers and Predictive Insights

ANALYSIS AND TECHNICAL REPORT

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1. EXECUTIVE SUMMARY

1.1 Overview

Food waste remains a significant global sustainability challenge, contributing to environmental degradation, resource inefficiency, and economic losses while undermining efforts to achieve the **United Nations Sustainable Development Goal (SDG) 12.3**.

This project aims to understand global food waste patterns and examine the relationship between food waste and selected socioeconomic and environmental factors across countries. Data from the **UNEP Food Waste Index Report 2024**, **World Bank datasets** (GDP per capita, population, population density, and income group), and **Trading Economics** temperature data were integrated into a consolidated master dataset covering 233 countries.

A structured methodology involving data extraction, cleaning, validation, country matching, dataset integration, exploratory data analysis, Tableau visualizations, and predictive modeling was applied to ensure data quality and analytical reliability. The analysis found that **households account for most of the global food waste**, while food waste levels vary across income groups and show stronger associations with certain socioeconomic factors than environmental factors. Multiple predictive models were evaluated, with machine learning approaches demonstrating improved predictive performance compared to baseline regression models in estimating household food waste.

Based on these findings, the study recommends **prioritizing household-focused waste reduction initiatives**, strengthening global food waste measurement and reporting standards, and incorporating additional socioeconomic and behavioral indicators to support future research and policymaking.

1.2 Key Highlights

This study analyzed food waste data from **233 countries** and integrated **11 external datasets** covering socioeconomic and environmental indicators to create a comprehensive global food waste dataset. A rigorous **Country Matching Audit** was conducted to standardize country references and ensure accurate integration across multiple data sources.

An interactive **Tableau dashboard suite** was developed to visualize global food waste patterns, sector contributions, and key relationships with demographic and economic factors. Statistical analysis and exploratory data visualization were used to uncover trends, correlations, and high-risk countries. Multiple predictive models, including Linear Regression and Random Forest were evaluated to identify the most effective approach for estimating household food waste.

2. INTRODUCTION

2.1 Background

Food waste is a growing global sustainability challenge that results in the loss of valuable resources, generates significant economic costs, and contributes to environmental issues such as greenhouse gas emissions and inefficient use of land, water, and energy. Recognizing its global impact, the United Nations included food waste reduction under **Sustainable Development Goal (SDG) 12.3, which aims to halve per capita food waste at the retail and consumer levels and reduce food losses across supply chains by 2030**. Understanding the factors associated with food waste is essential for supporting evidence-based policies and initiatives that promote more sustainable food systems.

2.2 Problem Statement

Despite increasing global attention on food waste, there is **limited understanding of how food waste levels vary across countries and the extent to which socioeconomic and environmental factors influence these differences**. Without such insights, policymakers and sustainability stakeholders may face challenges in designing targeted and effective strategies to reduce food waste and support the achievement of SDG 12.3.

2.3 Project Objectives

2.3.1 Primary Objectives

This project aims to understand global food waste patterns, identify key socioeconomic and environmental factors associated with food waste, compare food waste across sectors and income groups, and develop a predictive model for estimating household food waste.

2.3.2 Secondary Objectives

To support these objectives, a consolidated global food waste dataset was created by integrating multiple data sources, accompanied by interactive Tableau dashboards to visualize insights and actionable recommendations to support policymakers, sustainability practitioners, and researchers in reducing food waste.

2.4 Target Audience

By transforming fragmented global datasets into actionable insights, this analysis enables decision-makers, practitioners, researchers, and future analysts to better understand the drivers of food waste, prioritize interventions, and support more effective strategies for achieving sustainable food systems and reducing food waste worldwide.

Audience	Role as an Enabler
UNEP	This analysis provides evidence-based insights into global food waste patterns and drivers, enabling UNEP to strengthen international monitoring frameworks, prioritize interventions, and accelerate progress toward SDG 12.3 .
Policymakers and Government Agencies	This analysis identifies high-impact sectors and country-specific food waste challenges, enabling policymakers to design targeted regulations, incentives, and public awareness initiatives that maximize waste reduction outcomes .
Sustainability Practitioners	This analysis highlights where food waste reduction efforts can deliver the greatest impact, enabling sustainability practitioners to develop data-driven programs, partnerships, and performance measurement frameworks .
Researchers	This analysis offers a validated and integrated global dataset that enables deeper investigation into food waste drivers, supports the

Audience	Role as an Enabler
	development of advanced analytical models, and contributes to evidence-based sustainability research.
Non-Governmental Organizations (NGOs)	This analysis identifies priority populations and sectors for intervention, enabling NGOs to focus advocacy, education, and community engagement efforts where they can create the most meaningful and measurable impact.
Students and Data Analysts	This analysis demonstrates a complete end-to-end analytics workflow, enabling students and analysts to apply best practices in data integration, validation, visualization, and predictive modeling to real-world sustainability challenges.

2.5 Success Criteria

2.5.1 Analytical Success

Analytical success is achieved when the study **identifies the most influential socioeconomic and environmental factors** associated with food waste, **uncovered meaningful global patterns** across countries and income groups, and **generates insights** that deepen understanding of food waste behaviors and trends. The findings should provide a clear, evidence-based narrative that supports further investigation and informed decision-making.

2.5.2 Technical Success

Technical success is achieved by developing a **clean, validated, and integrated master dataset** from multiple sources, ensuring data quality, consistency, and reliability throughout the analysis. Success also includes building a **robust predictive regression model with meaningful explanatory power** and delivering **interactive Tableau dashboards that effectively communicate insights** through data visualization.

2.5.3 Business & Sustainability Success

Business and sustainability success is achieved when the analysis translates data into **actionable recommendations that support food waste reduction efforts and sustainability goals**. The project should enable evidence-based decision-making for policymakers and stakeholders while increasing awareness of the environmental, social, and economic impacts of food waste at a global scale.

3. RESEARCH METHODOLOGY

3.1 Project Framework

This project followed a structured end-to-end analytics framework, from data acquisition and preparation through analysis and predictive modeling, to transform multiple datasets into actionable insights and recommendations for addressing global food waste.

- a. Data Collection
- b. Data Loading
- c. Data Cleaning
- d. Data Validation
- e. Country Matching Audit
- f. Master Dataset Creation
- g. Exploratory Data Analysis
- h. Tableau Visualization
- i. Predictive Modeling
- j. Recommendations

3.2 Tools and Technologies

Below are the tools and technologies used throughout this project.

Tool	Purpose
Python	Data Processing
Pandas	Data Transformation
NumPy	Statistical Analysis
Scikit-Learn	Predictive Modeling
Jupyter Notebook	Python Code Development
Tableau	Data Visualization
GitHub	Version Control & Project Documentation
Trello	Project Management

4. DATA SOURCES

4.1 UNEP Food Waste Index Report 2024

Source: <https://www.unep.org/resources/publication/food-waste-index-report-2024>

Data availability: Downloadable report in pdf format

Filename: food_waste_index_report_appendix_2024.pdf

Dataset	Pages	# of rows
Household Estimates	29-36	233
Food Service Estimates	37-47	233
Retail Estimates	47-55	233

4.2 World Bank Datasets

Source: <https://data.worldbank.org>

Data availability: Downloadable files in csv format

Dataset	Filename	# of rows
GDP Per Capita	API_NY.GDP.PCAP.CD_DS2_en_csv_v2_121663.csv	266
Population Density	API_EN.POP.DNST_DS2_en_csv_v2_1453.csv	266
Tourism Arrivals	API_ST.INT.ARVL_DS2_en_csv_v2_126984.csv	266
Income Group	CLASS_2025_10_07.xlsx	218
CO ₂ Emissions	API_EN.GHG.CO2.WA.MT.CE.AR5_DS2_en_csv_v2_12756.csv	266
Access to Electricity	API_EG.ELC.ACCS.ZS_DS2_en_csv_v2_127016.csv	266
Population Total	API_SP.POP.TOTL_DS2_en_csv_v2_127039.csv	266

4.3 Trading Economics

Source: <https://tradingeconomics.com/country-list/temperature>

Data availability: Copied the table from the website and saved it as a csv

Dataset	Filename	# of rows
Average Temperature by Country	Average Temperature By Country 2024.csv	205

4.4 Data Dictionary

Source Filename: master_dataset_global_food_waste.csv

Field Name	Description	Datatype
country_unep	Standardized country name based on the UNEP Food Waste Index Report 2024 and used as the master country key.	str

Field Name	Description	Datatype
household_year	Reporting year for household food waste estimates.	int64
household_region	UNEP regional classification of the country.	str
household_kg_per_capita_year	Annual household food waste generated per person (kg/capita/year).	int64
household_tonnes_year	Annual household food waste generated at country level (tonnes/year).	float64
household_confidence	UNEP confidence level assigned to the household food waste estimate.	str
food_service_year	Reporting year for food service food waste estimates.	int64
food_service_kg_per_capita_year	Annual food service food waste generated per person (kg/capita/year).	int64
food_service_tonnes_year	Annual food service food waste generated at country level (tonnes/year).	float64
food_service_confidence	UNEP confidence level assigned to the food service food waste estimate.	str
retail_year	Reporting year for retail food waste estimates.	int64
retail_kg_per_capita_year	Annual retail food waste generated per person (kg/capita/year).	int64
retail_tonnes_year	Annual retail food waste generated at country level (tonnes/year).	float64
retail_confidence	UNEP confidence level assigned to the retail food waste estimate.	str
average_temperature	Average annual temperature of the country.	float64
temperature_year	Reference year of the temperature measurement.	float64
temperature_unit	Unit of measurement for temperature.	str
income_year	Reference year of the World Bank income classification dataset.	float64
income_region	World Bank regional classification.	str
income_group	World Bank income group classification.	str
income_lending_category	World Bank lending eligibility classification.	str
gdp_per_capita_year	Latest available year of GDP per capita data.	float64
gdp_per_capita	Gross Domestic Product per capita (current US\$).	float64
population_density_year	Latest available year of population density data.	float64
population_density	Number of people per square kilometre of land area.	float64
tourism_arrivals_year	Latest available year of international tourism arrivals data.	float64
tourism_arrivals	Number of international tourist arrivals.	float64
access_to_electricity_year	Latest available year of access to electricity data.	float64
access_to_electricity	Percentage of population with access to electricity.	float64
total_population_year	Latest available year of population data.	float64
total_population	Total population of the country.	float64

Field Name	Description	Datatype
total_food_waste_kg_per_capita_year	Total food waste generated across household, food service and retail sectors per person (kg/capita/year).	int64
total_food_waste_tonnes_year	Total food waste generated across all sectors at country level (tonnes/year).	float64
household_share_pct	Percentage contribution of household food waste to total food waste.	float64
food_service_share_pct	Percentage contribution of food service food waste to total food waste.	float64
retail_share_pct	Percentage contribution of retail food waste to total food waste.	float64
main_contributing_sector	Food waste sector contributing the largest share of total food waste.	str
tourists_per_resident	Ratio of international tourist arrivals to resident population.	float64
income_rank	Numeric ranking assigned to income groups for analytical purposes.	float64
population_size_category	Population size category derived from total population values.	str
temperature_band	Temperature classification band derived from average annual temperature.	str
gdp_per_capita_category	GDP per capita category derived from GDP per capita values.	str
coverage_score	Count of available non-null analytical variables for each country.	int64
sector_share_total	Validation field representing the combined sector contribution percentages.	float64

5. DATA LOADING AND ACQUISITION

5.1 UNEP Data Extraction Process

5.1.1 Household Estimates

The Household Estimates dataset was extracted from the UNEP Food Waste Index Report 2024 and underwent a series of data preparation steps to address extraction challenges commonly associated with PDF-based tables.

Key corrections were conducted to ensure the accuracy and consistency of the final 233-country dataset:

- resolving header formatting inconsistencies,
- fixing column alignment issues,
- and validating M49 country codes.

Additional transformations were performed to standardize country information, remove extraction anomalies, and align the dataset structure with the requirements of the subsequent data validation and integration processes.

5.1.2 Food Service Estimates

The Food Service Estimates dataset was extracted from the UNEP Food Waste Index Report 2024 and presented significant challenges due to PDF extraction limitations such as:

- multi-line country entries,
- fragmented records,
- and missing confidence values.

As these formatting inconsistencies could not be fully resolved through standard data transformation techniques without extensive manual coding, a Large Language Model (LLM) was used to extract the entire Food Service Estimates section directly from the PDF into a structured CSV format.

To ensure data accuracy and integrity, the extracted dataset was validated through a record-by-record comparison against the original UNEP source tables, achieving a 100% match across all 233 country records.

5.1.3 Retail Estimates

The Retail Estimates dataset was extracted from the UNEP Food Waste Index Report 2024 using a multi-table extraction process, as the source data was distributed across several tables spanning multiple pages.

Data preparation focused on:

- removing repeated headers,
- standardizing column structures,
- and consolidating the extracted tables into a single consistent dataset.

These steps ensured the completeness, accuracy, and comparability of the final dataset covering 233 countries.

5.2 World Bank Data Loading

Multiple datasets were obtained from the World Bank to provide socioeconomic, demographic, environmental, and infrastructure context for the food waste analysis, including:

- GDP per Capita
- Total Population
- Population Density
- Income Group Classification
- International Tourism Arrivals
- CO₂ Emissions, and
- Access to Electricity

To ensure consistency and relevance, **only data from 2015 to 2024 was retained**, country identifiers and data types were standardized, and key metadata was preserved to support accurate integration, validation, and cross-country comparison throughout the analysis.

5.3 Trading Economics Data Loading

The Average Temperature by Country dataset was sourced from Trading Economics by exporting the published table into a CSV file and loading it into a Pandas DataFrame for transformation and analysis. The dataset underwent standardization of country identifiers, reporting dates, data types, and supporting metadata to ensure consistency with the other datasets and enable seamless integration into the master dataset.

5.4 Data Loading Summary

For all the datasets sourced from multiple organizations (UNEP, World Bank, and Trading Economics), the loading process involved:

- standardizing reporting periods,
- transforming source-specific structures into analysis-ready formats,
- harmonizing country identifiers,
- validating data types, and
- retaining key metadata attributes

These activities established a consistent foundation for the subsequent data cleaning, validation, integration, and analytical phases of the project.

6. DATA CLEANING AND PREPARATION

6.1 Data Quality Assessment

A structured data quality assessment was performed to evaluate the completeness, uniqueness, validity, and consistency of all source datasets. The assessment focused on:

- identifying missing values,
- duplicate records,
- invalid entries,
- and formatting inconsistencies, including country identifiers, data types, units of measure, and reporting periods

The findings informed the data cleaning and preparation activities, ensuring the accuracy and reliability of the final master dataset used for analysis and predictive modeling.

6.2 UNEP Dataset Cleaning

6.2.1 Household Estimates

The Household Estimates dataset underwent extensive cleaning to address extraction-related issues, including header formatting inconsistencies, column misalignments, and country field variations. Country names were standardized, M49 country codes were validated against the source data, and structural corrections were applied to ensure the accuracy and consistency of the final 233-country dataset.

6.2.2 Food Service Estimates

The Food Service Estimates dataset required the consolidation of fragmented country records and recovery of missing confidence values caused by PDF extraction limitations. As these issues could not be efficiently resolved through standard data transformation techniques, an **AI-assisted approach** was used to extract the entire Food Service Estimates section directly into a structured CSV format. The resulting dataset was validated through a record-by-record comparison with the original UNEP source tables, achieving a **100% match across all 233 countries**.

6.2.3 Retail Estimates

The Retail Estimates dataset was cleaned by removing repeated headers, consolidating data extracted from multiple source tables, and standardizing the dataset structure. These steps ensured a consistent format and improved comparability across countries and food waste sectors for subsequent analysis.

6.3 World Bank Data Cleaning

The World Bank datasets were cleaned through a series of standardization and validation procedures, including the selection of analysis-relevant observations from 2015–2024, country name harmonization, data type validation, handling of missing and inconsistent records, and structural normalization.

This ensured a consistent and reliable set of socioeconomic, demographic, environmental, and infrastructure indicators for integration with the food waste dataset and subsequent analytical activities.

6.4 Trading Economics Data Cleaning

The Trading Economics Average Temperature by Country dataset was cleaned and standardized to improve consistency and comparability across countries. Data preparation activities included:

- transforming reference dates into a standardized reporting format,
- validating and standardizing temperature values,
- and harmonizing units of measure to a consistent Celsius format

These steps ensured the dataset was analysis-ready and could be accurately integrated with the other socioeconomic and environmental datasets in the master dataset.

7. DATA VALIDATION AND QUALITY ASSURANCE

7.1 Validation Framework

A structured validation framework was implemented to ensure the accuracy, completeness, and reliability of all datasets used in the analysis. Validation activities included:

- **record count validation** to confirm completeness against source data,
- **uniqueness validation** to identify duplicate records,
- **missing value validation** to assess data completeness,
- **source-to-output validation** to verify that extracted and transformed data accurately reflected the original sources,
- and **data consistency validation** to ensure standardized country identifiers, reporting periods, units of measure, and data formats across datasets

These quality assurance measures provided confidence in the integrity of the final master dataset and the validity of the subsequent analysis and predictive modeling results.

7.2 Data Quality Outcomes

The data quality assessment identified a small number of issues related to:

- PDF extraction limitations,
- formatting inconsistencies,
- country naming variations,
- missing values in selected source fields,
- and differences in dataset coverage across countries and reporting periods

These issues were addressed through data cleaning, standardization, validation checks, country matching, structural corrections, and source-to-output verification, including a fully validated AI-assisted extraction of the UNEP Food Service Estimates dataset. As a result, all eleven datasets successfully passed validation with no critical data quality issues identified, providing a reliable and analysis-ready foundation for the master dataset, exploratory analysis, Tableau dashboards, and predictive modeling, with only one minor source exception retained to preserve data lineage in the Trading Economics temperature dataset.

7.3 Validation Results

SNo	Dataset	Data Source	Reference Period (in years)	Key Field	# of Rows (Raw)	# of Unique Countries	# of Rows (Transformed)	Missing Values (Field-Level)
1	Household Food Estimates	UNEP Food Waste Index Report 2024	2024	country	233	233	233	5-household_tonnes_year
2	Food Service Food Estimates	UNEP Food Waste Index Report 2024	2024	country	233	233	233	5-food_service_tonnes_year
3	Retail Food Estimates	UNEP Food Waste Index Report 2024	2024	country	233	233	233	5-retail_tonnes_year
4	Average Temperature by Country	Trading Economics	2024	country	205	205	205	None
5	GDP Per Capita	World Bank Open Data	2015-2024	country_name	266	261	2,560 ¹	None
6	Population Density	World Bank Open Data	2015-2023	country_name	266	264	2,326 ¹	None
7	Tourism Arrivals	World Bank Open Data	2015-2020	country_name	266	242	1,299 ¹	None
8	Income Group	World Bank Open Data	2024	economy	218	218	218	2-income_group 73-lending_category
9	CO ₂ Emissions	World Bank Open Data	2015-2024	country_name	266	148	1,476 ¹	None
10	Access to Electricity	World Bank Open Data	2015-2023	country_name	266	263	2,367 ¹	None
11	Total Population	World Bank Open Data	2015-2024	country_name	266	265	2,650 ¹	None

¹For World Bank Datasets, transformed rows represent country-year observations after reshaping from wide to long format.

SNo	Dataset	Extraction Methodology	Data Quality Issues	Source Coverage Limitations	Resolution	CSV Filename	Validation Status
1	Household Food Estimates	Table extraction in PDF via tabula	-	-	N.A.	household_estimates_cleaned.csv	Fully Validated
2	Food Service Food Estimates	Table extraction in PDF via LLM	-	-	N.A.	food_service_estimates_cleaned.csv	Fully Validated
3	Retail Food Estimates	Table extraction in PDF via tabula	-	-	N.A.	retail_estimates_cleaned.csv	Fully Validated
4	Average Temperature by Country	Read CSV file via pandas	Isle of Man reference period = Dec 2021		Retained source reference date (2021-12-31)	temperature_country_cleaned.csv	Validated with minor exception
5	GDP Per Capita	Read CSV file via pandas	-	-	N.A.	gdp_cleaned.csv	Fully Validated
6	Population Density	Read CSV file via pandas	-	Latest available data is 2023	N.A.	population_density_cleaned.csv	Fully Validated
7	Tourism Arrivals	Read CSV file via pandas	-	Latest available data is 2020 (pre-pandemic period)	N.A.	tourism_arrival_cleaned.csv	Fully Validated
8	Income Group	Read XLSX file via openpyxl	-	-	N.A.	income_group_cleaned.csv	Fully Validated
9	CO ₂ Emissions	Read CSV file via pandas	-	-	N.A.	co2_emissions_cleaned.csv	Fully Validated
10	Access to Electricity	Read CSV file via pandas	-	Latest available data is 2023	N.A.	access_to_electricity_cleaned.csv	Fully Validated
11	Total Population	Read CSV file via pandas	-		N.A.	total_population_cleaned.csv	Fully Validated

8. COUNTRY MATCHING AUDIT

8.1 Purpose of Country Matching

The datasets used in this study originated from multiple sources, each employing different country naming conventions and reference standards, which created challenges for accurate data integration and analysis.

Examples included variations such as **"Bahamas"** and **"Bahamas, The"**, **"Türkiye"** and **"Turkey"**, and **"Bolivia (Plurin. State of)"** and **"Bolivia"**, which refer to the same countries but are recorded differently across datasets.

The Country Matching Audit was therefore conducted to standardize country identifiers, ensure accurate dataset integration, and establish a consistent country reference framework for the master dataset.

8.2 UNEP as Master Reference

The **UNEP Food Waste Index Report 2024** was selected as the master reference dataset because it served as the core analytical foundation of the study, provided the largest and most comprehensive country coverage among the food waste sources, and contained the primary food waste metrics being analyzed.

As the authoritative source for household, food service, and retail food waste estimates, UNEP was treated as the project's **source of truth**, with all external datasets aligned to its country definitions and coverage wherever possible.

8.3 Matching Methodology

A structured country matching methodology was applied to ensure consistent integration across all datasets.

Countries were first matched through **direct matches** where names aligned exactly, followed by **country name standardization** using a mapping framework to reconcile differences in naming conventions across sources. **Exception handling** was applied to countries with incomplete coverage or no corresponding match in the UNEP reference dataset.

8.4 Country Matching Results

The Country Matching Audit successfully aligned countries across the UNEP, World Bank, and Trading Economics datasets through a combination of direct matches, country name standardization, and manual mapping.

Most countries were matched directly, while a smaller number required mapping updates to reconcile differences in naming conventions across sources.

The audit also identified countries that were available only in the UNEP dataset and others that existed only in external datasets, ensuring that all coverage differences were documented and appropriately managed during the creation of the master dataset.

Matching Result	Examples	Filename
Direct Matches	Australia, Costa Rica, Singapore	audit_direct_match_updates.csv
Country Mapping	Turkey → Türkiye Bahamas, The → Bahamas Bolivia → Bolivia (Plurin. State of)	audit_country_mapping_updates.csv

Matching Result	Examples	Filename
Countries in UNEP only	Anguilla, Martinique, Wallis and Futuna Islands	audit_country_found_only_in_unep.csv
Countries not in UNEP	Countries: • Channel Islands • Kosovo • Taiwan, China • West Bank and Gaza Non-country Aggregates: • Arab World • High income • European Union • OECD members	audit_country_not_found_in_unep.csv

8.5 Audit Summary

The Country Matching Audit achieved a high degree of alignment across all source datasets, with most countries matched directly and additional countries successfully integrated through standardized mapping of naming convention differences. A small number of countries were excluded from integration due to differences in dataset coverage or the absence of corresponding records in the UNEP reference dataset.

The audit provided a transparent and consistent country reference framework that strengthened the accuracy, completeness, and reliability of the final master dataset.

9. MASTER DATASET DEVELOPMENT

9.1 Data Integration Strategy

A master dataset was developed by **integrating 10 datasets from UNEP, World Bank, and Trading Economics using the UNEP Food Waste Index Report 2024 as the primary reference dataset.**

Following data cleaning, validation, and the Country Matching Audit, datasets were merged using standardized country identifiers to combine food waste, socioeconomic, demographic, environmental, infrastructure, and tourism indicators into a single analytical dataset.

Although **CO₂ Emissions** was initially considered for inclusion, **it was excluded** from the final master dataset due to a **low country matching rate of 42.9%** identified during the integration process, which would have significantly reduced dataset coverage and analytical reliability. This approach maximized country coverage while maintaining the integrity and consistency of the final dataset used for exploratory analysis, dashboard development, and predictive modeling.

	DataSource	Direct Match	Naming Convention Match	Found Only in UNEP	Not Found in UNEP	Matched Countries	# of Countries (Source)	Coverage %
0	Access to Electricity	183	30	20	50	213	263	91.4
1	CO2 Emissions	84	16	133	48	100	148	42.9
2	Food Service Estimates	233	0	0	0	233	233	100.0
3	GDP per Capita	181	29	23	51	210	261	90.1
4	Income Group	184	30	19	4	214	218	91.8
5	Population Density	184	30	19	50	214	264	91.8
6	Retail Estimates	233	0	0	0	233	233	100.0
7	Temperature	177	24	32	4	201	205	86.3
8	Total Population	184	30	19	51	214	265	91.8
9	Tourism Arrivals	168	27	38	47	195	242	83.7

Table 1. Country Matching Summary

9.2 Master Dataset Structure

9.2.1 Key Variable

- Country

9.2.2 Food Waste Metrics & Attributes

Household	Food Service	Retail
• Household Kg/Capita • Household Tonnes • Household Year • Household Region • Household Confidence	• Food Service Kg/Capita • Food Service Tonnes • Food Service Year • Food Service Confidence	• Retail Kg/Capita • Retail Tonnes • Retail Year • Retail Confidence

9.2.3 Socioeconomic Variables

- Income Group, Year, Region and Lending Category
- GDP Per Capita and Year
- Total Population and Year
- Population Density and Year
- Tourism Arrivals and Year
- Access to Electricity and Year

9.2.4 Environmental Variables

- Average Temperature, Year, and Unit of Temperature

9.2.5 Additional Metrics

- Total Food Waste Kg per Capita per Year
- Total Food Waste Tonnes per Year
- Household Share Percentage
- Food Service Share Percentage
- Retail Share Percentage
- Main Contributing Sector
- Tourists per Resident
- Income Group Rank
- Population Size Category
- Temperature Band
- GDP per Capita Category
- Coverage Score
- Sector Share Total

9.3 Final Dataset Profile

The final master dataset consists of **233 country-level records** and **44 variables** encompassing food waste, socioeconomic, demographic, environmental, infrastructure, and tourism indicators.

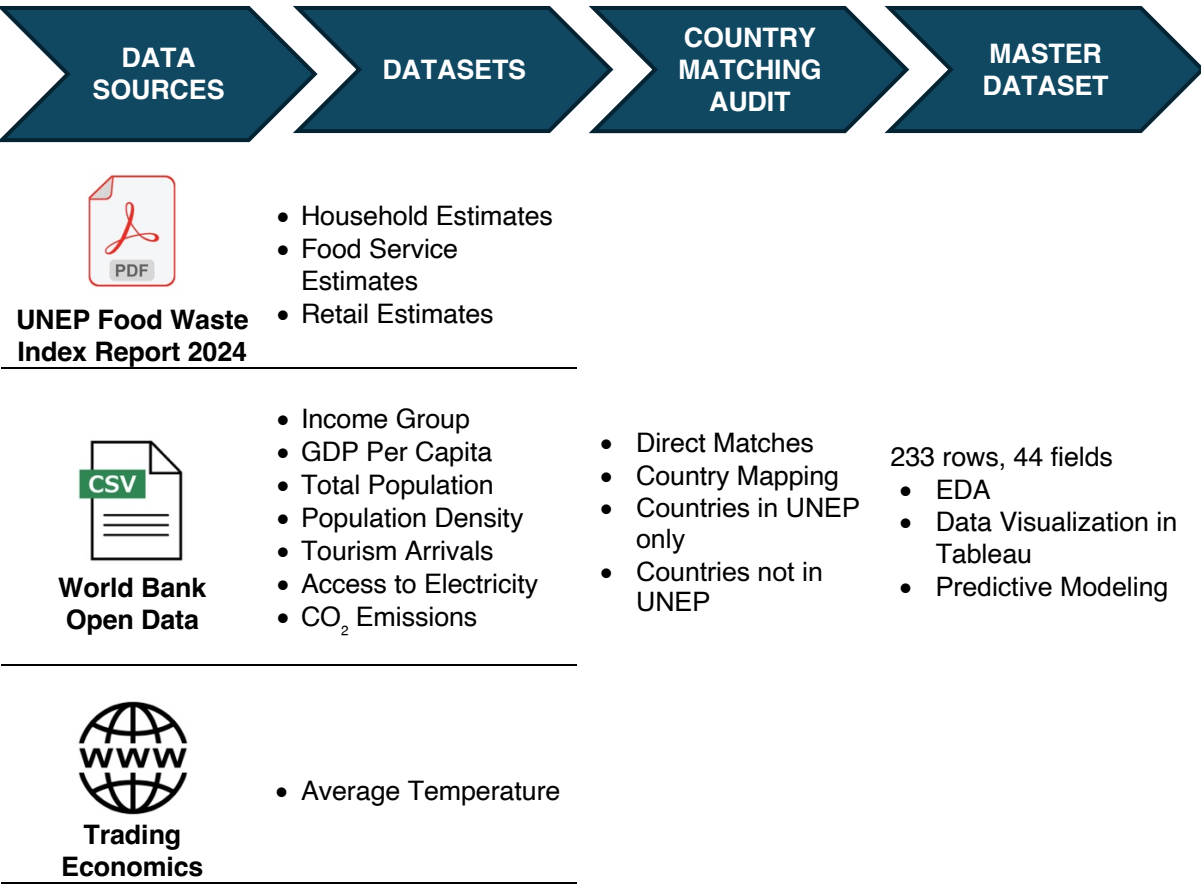
Following data integration, cleaning, validation, and country matching activities, the dataset achieved **high coverage of at least 83%** across the included data sources, with missing values limited to countries where source data was unavailable.

Income group classifications were unavailable for 21 countries in the master dataset. Consequently, analyses requiring income group segmentation were conducted on 212 countries with valid income group assignments, representing 91.0% of the total 233-country dataset. Countries with

missing income group information were excluded only from income-group-based visualisations and statistical summaries.

The resulting dataset provides a comprehensive and reliable global view of food waste and its potential drivers, forming a robust foundation for exploratory analysis, interactive dashboards, and predictive modeling.

9.4 Data Model Diagram



10. EXPLORATORY DATA ANALYSIS (EDA)

10.1 Chart 1: Top 10 Contributors to Global Food Waste

Top 10 Contributors to Global Food Waste

China and India together account for approximately one-third of food waste generated among the top contributing countries

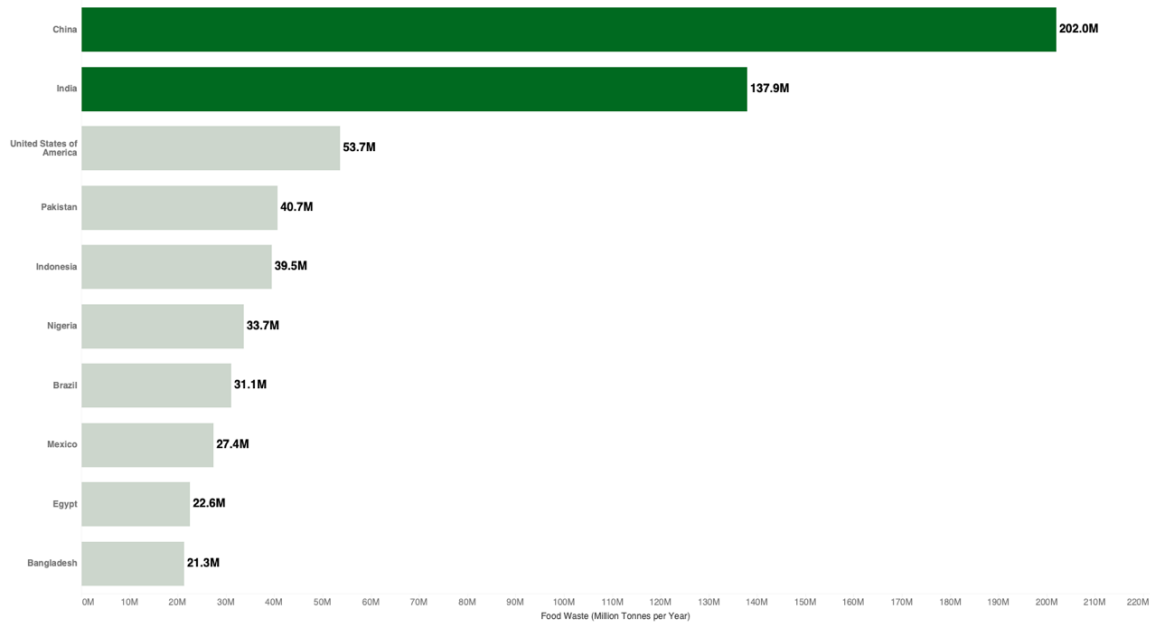


Figure 1. Top 10 Contributors to Global Food Waste

10.1.1 Purpose

This chart identifies the **countries generating the largest volumes of food waste globally** and highlights where the greatest opportunities for waste reduction may exist. It also provides an initial view of the relationship between population size and total food waste generation.

10.1.2 Methodology

Countries were **ranked using the total annual food waste metric**, calculated as the combined household, food service, and retail food waste estimates from the UNEP Food Waste Index. The top 10 countries were then visualized in Tableau using a **horizontal bar chart** sorted in descending order.

10.1.3 Key Findings

China, India, and the United States are the largest contributors to global food waste by volume, with the top 10 countries largely consisting of highly populated nations. This suggests that population size is a major driver of total food waste generation, although it does not necessarily correspond to higher food waste per person.

10.1.4 Validation

The Tableau results were independently validated in Python by **sorting countries according to total annual food waste and comparing the rankings with the visualization output**. The validation confirmed that the countries and values displayed in the chart accurately reflected the underlying dataset.

10.1.5 Business / Sustainability Implication

Because a significant share of global food waste originates from a small number of highly populated countries, targeted interventions in these nations could deliver the largest reductions in total food waste volumes. This highlights the **importance of population-scale policies, infrastructure**

improvements, and consumer awareness initiatives in achieving global food waste reduction goals.

10.2 Chart 2: Countries Generating the Highest Food Waste per Person

Countries Generating the Highest Food Waste per Person

Countries with the highest food waste intensity are not necessarily the most populous or wealthiest nations

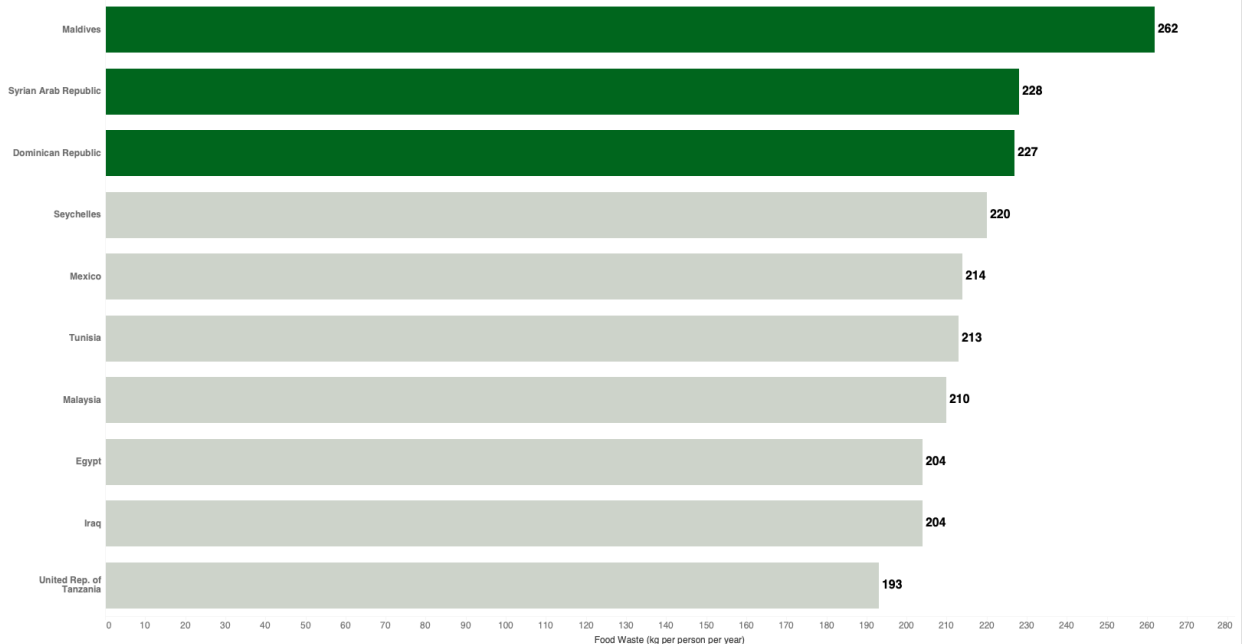


Figure 2. Countries Generating the Highest Food Waste per Person

10.2.1 Purpose

This chart identifies the **countries with the highest food waste intensity**, measured as annual food waste generated per person (kg/person/year). It aims to determine whether countries with the highest per-capita food waste differ from those contributing the largest total food waste volumes.

10.2.2 Methodology

Total food waste per capita was calculated by combining household, food service, and retail food waste estimates from the UNEP Food Waste Index. Countries were ranked in descending order, and the top ten countries with the highest food waste per person were visualized using a **horizontal bar chart**.

10.2.3 Key Findings

The **Maldives recorded the highest food waste intensity at 262 kg/person/year**, followed by the Syrian Arab Republic (228 kg/person/year) and the Dominican Republic (227 kg/person/year). Notably, none of the world's most populous countries appear in the top rankings, suggesting that food waste intensity and total food waste volume are influenced by different factors.

10.2.4 Validation

The **ranking and food waste per capita values were independently validated in Python using the master dataset and matched the Tableau visualization results**. Validation confirmed that the displayed countries and values accurately reflect the underlying UNEP food waste estimates.

10.2.5 Business / Sustainability Implication

The results indicate that food waste challenges are not limited to large economies or highly populated countries; smaller nations can also exhibit very high levels of waste per person. This suggests that **targeted interventions addressing consumer behavior, tourism activity, food**

consumption patterns, and local waste management practices may be more effective than population-based approaches alone.

10.3 Chart 3: Income vs Food Waste Intensity

Upper-Middle-Income Countries Record the Highest Food Waste Per Person

Population-weighted food waste intensity reveals that income level alone does not explain food waste generation patterns.

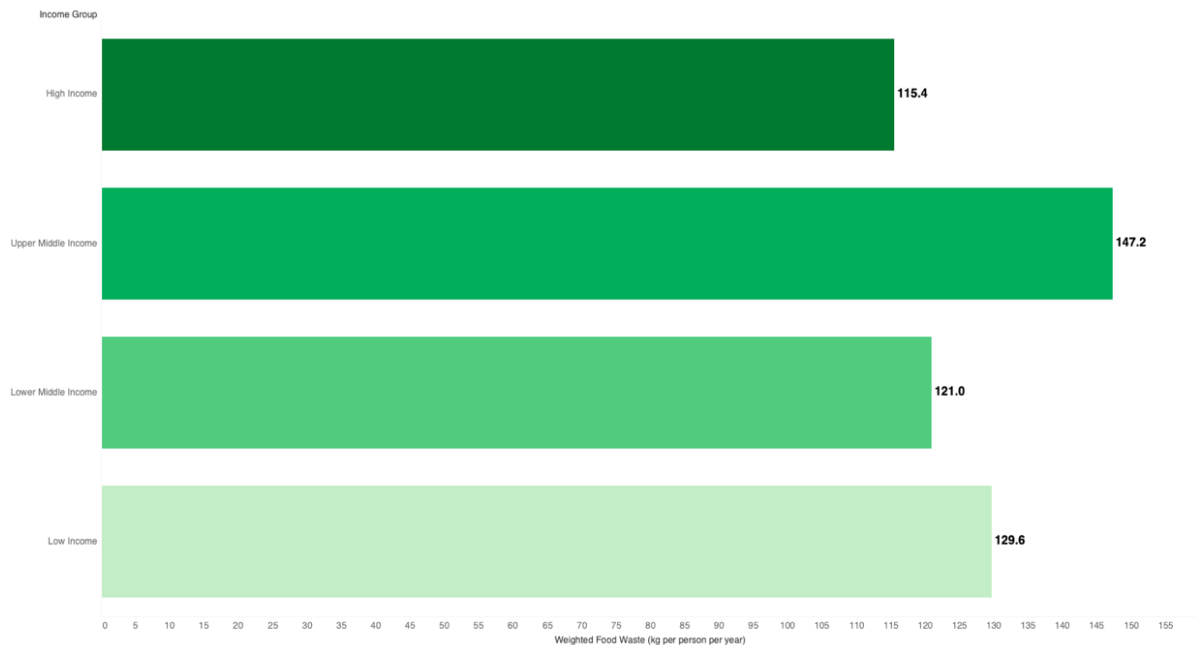


Figure 3. Income vs Food Waste Intensity

10.3.1 Purpose

This chart examines whether **food waste intensity varies across World Bank income groups by comparing population-weighted average food waste per person**. The analysis seeks to determine whether higher income levels are consistently associated with greater food waste generation.

10.3.2 Methodology

Countries were grouped according to World Bank income classifications (Low, Lower Middle, Upper Middle, and High Income). A **population-weighted average food waste per person was calculated for each group using combined household, food service, and retail food waste estimates** from the UNEP Food Waste Index.

10.3.3 Key Findings

Upper-middle-income countries recorded the highest weighted average food waste intensity at 147.2 kg/person/year, followed by low-income countries at 129.6 kg/person/year, while high-income countries recorded the lowest average at 115.4 kg/person/year. The results indicate that **food waste intensity does not increase uniformly with income and may be influenced by broader factors beyond economic status alone**.

10.3.4 Validation

The weighted averages displayed in Tableau were independently validated using Python calculations based on the master dataset and matched the reported values. Additional validation using median food waste values produced a similar ranking pattern, confirming that the findings were not driven by a small number of extreme observations.

10.3.5 Business / Sustainability Implication

The findings suggest that income level alone is not a reliable predictor of food waste intensity, highlighting the importance of considering factors such as consumption behavior, tourism activity, food supply chain efficiency, and waste management practices. **Effective food waste reduction strategies should therefore be tailored to country-specific conditions rather than relying solely on economic development status.**

10.4 Chart 4: Population vs Food Waste

Population Size Is a Strong Predictor of Total Food Waste Generation
Population size is strongly associated with total food waste generation across countries ($R^2 \approx 0.89$).

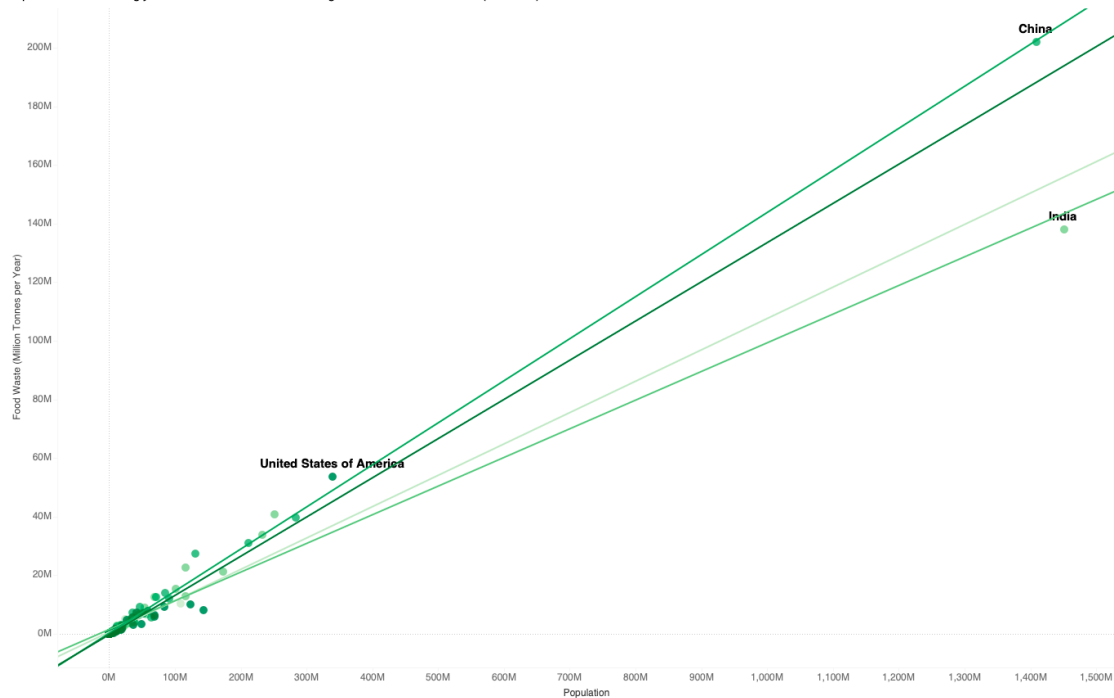


Figure 4. Population vs Food Waste (All Countries)

The Population–Food Waste Relationship Persists Beyond China and India

Population remains a strong predictor of food waste generation even after excluding China and India.



Figure 5. Population vs Food Waste (excluding China and India)

10.4.1 Purpose

These charts examine the **relationship between national population size and total food waste generation** and assess whether larger populations consistently generate greater food waste volumes. Chart 4B further tests the robustness of this relationship by excluding China and India, the two largest population outliers.

10.4.2 Methodology

Scatter plots were created using total population as the independent variable and total food waste (tonnes/year) as the dependent variable. **Trend lines were added by income group**, and statistical strength was evaluated using R^2 and p-values, both with and without China and India included in the analysis.

10.4.3 Key Findings

A strong positive relationship exists between population size and total food waste generation across all income groups, with R^2 values ranging from approximately 0.89 to 0.97 and p-values below 0.0001. The relationship remains strong even after excluding China and India, demonstrating that the observed pattern reflects a broad global trend rather than being driven solely by a small number of highly populated countries.

10.4.4 Validation

The population and food waste values displayed in Tableau were independently validated using Python and matched the master dataset. **The consistently high R^2 values across both analyses confirm that population size is a statistically significant predictor of total food waste generation.**

10.4.5 Business / Sustainability Implication

The findings indicate that population size is one of the strongest drivers of total food waste generation worldwide, explaining a substantial proportion of the variation observed across countries. While highly populated countries such as China, India, and the United States contribute significantly to global food waste, the persistence of the relationship after removing major outliers

suggests that **scalable waste reduction strategies are needed across both large and medium-sized population centres.**

10.5 Chart 5: Global Waste by Sector

Households Generate 60% of Global Food Waste

Household food waste alone exceeds the combined contribution of food service and retail sectors worldwide.

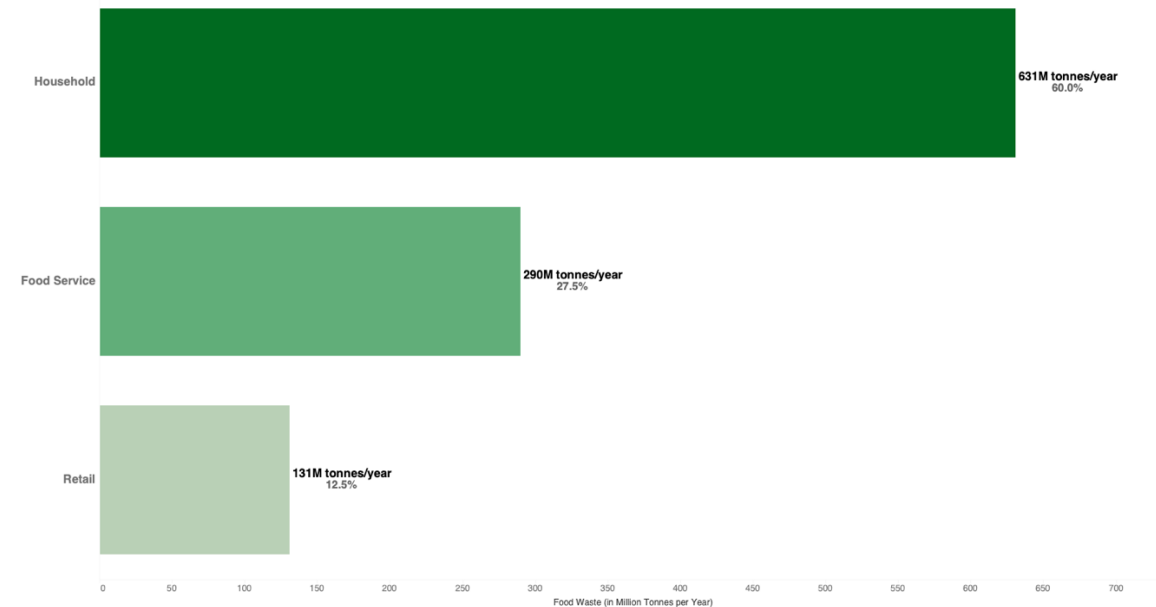


Figure 6. Global Waste by Sector

10.5.1 Purpose

This chart **compares the contribution of the household, food service, and retail sectors to total global food waste.** The objective is to identify which sector generates the largest share of food waste and where reduction efforts may have the greatest impact.

10.5.2 Methodology

Food waste tonnage from the UNEP Food Waste Index was aggregated across all 233 countries for each sector. **Sector totals were then converted into percentage shares of global food waste** and visualized using a **ranked horizontal bar chart.**

10.5.3 Key Findings

Households account for approximately **631 million tonnes of food waste annually**, representing **60.0%** of total global food waste, exceeding the combined contribution of the food service and retail sectors. Food service contributes **290 million tonnes (27.5%)**, while retail contributes **131 million tonnes (12.5%)**, highlighting households as the dominant source of food waste worldwide.

10.5.4 Validation

Sector totals were independently validated in Python using the master dataset, producing results of 630.96 million tonnes (Household), 289.78 million tonnes (Food Service), and 131.22 million tonnes (Retail). **Percentage shares calculated from the validated totals matched the values displayed in Tableau, confirming the accuracy of the visualization.**

10.5.5 Business / Sustainability Implication

The findings suggest that reducing household food waste should be the highest priority for governments, policymakers, and sustainability organizations, as it represents the largest opportunity for impact. While interventions across food service and retail remain important, **consumer education, food planning, storage practices, and behavioral change initiatives are likely to deliver the greatest reductions in global food waste and associated environmental impacts.**

10.6 Chart 6: Food Waste by Income Group and Sectors

Food Waste Is Concentrated in Middle Income Economies

Upper Middle Income and Lower Middle Income countries account for nearly 80% of total food waste, while Household Waste remains the dominant sector across all income groups

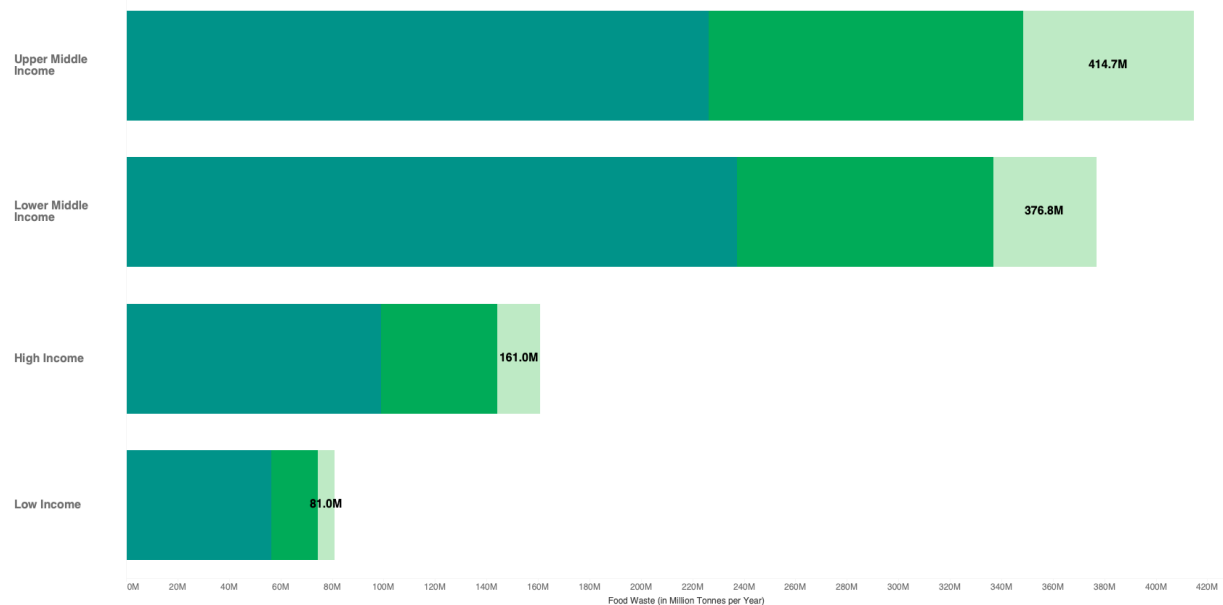


Figure 7. Food Waste by Income Group and Sectors

10.6.1 Purpose

This chart examines **how total food waste is distributed across income groups and identifies the relative contribution of the Household, Food Service, and Retail sectors within each group.** The objective is to determine whether food waste generation is concentrated in specific economic segments and to understand the sectoral drivers behind these patterns.

10.6.2 Methodology

Countries were grouped according to World Bank income classifications (High, Upper Middle, Lower Middle, and Low Income). **Total annual food waste (tonnes/year) from the UNEP Food Waste Index was aggregated by income group and segmented into Household, Food Service, and Retail waste using a stacked bar chart.**

10.6.3 Key Findings

Upper Middle Income countries generate the largest volume of food waste (414.7 million tonnes/year), followed closely by Lower Middle Income countries (376.8 million tonnes/year), together accounting for nearly 80% of total food waste. Across all income groups, Household waste is the dominant contributor, indicating that household consumption and disposal behaviours are the primary source of food waste regardless of economic status.

10.6.4 Validation

The analysis was performed using **212 countries with valid World Bank income group classifications; 21 countries with missing income group data were excluded**. Aggregate food waste totals were cross-checked against the master dataset to ensure accurate sector and income-group allocations.

10.6.5 Business / Sustainability Implication

The concentration of food waste in middle-income economies suggests that these countries offer the greatest opportunity for large-scale food waste reduction initiatives and sustainability interventions. Since Household waste consistently represents the largest share across all income groups, **consumer awareness campaigns, behavioural change programs, and household food management strategies are likely to deliver the highest impact globally**.

10.7 Chart 7: Food Waste Distribution by Income Group

Food Waste Varies Significantly Within Each Income Group

Upper-middle-income countries record the highest median food waste intensity (142 kg/person/year), while wide distributions and outliers across all income groups suggest that income alone does not fully explain food waste behaviour.

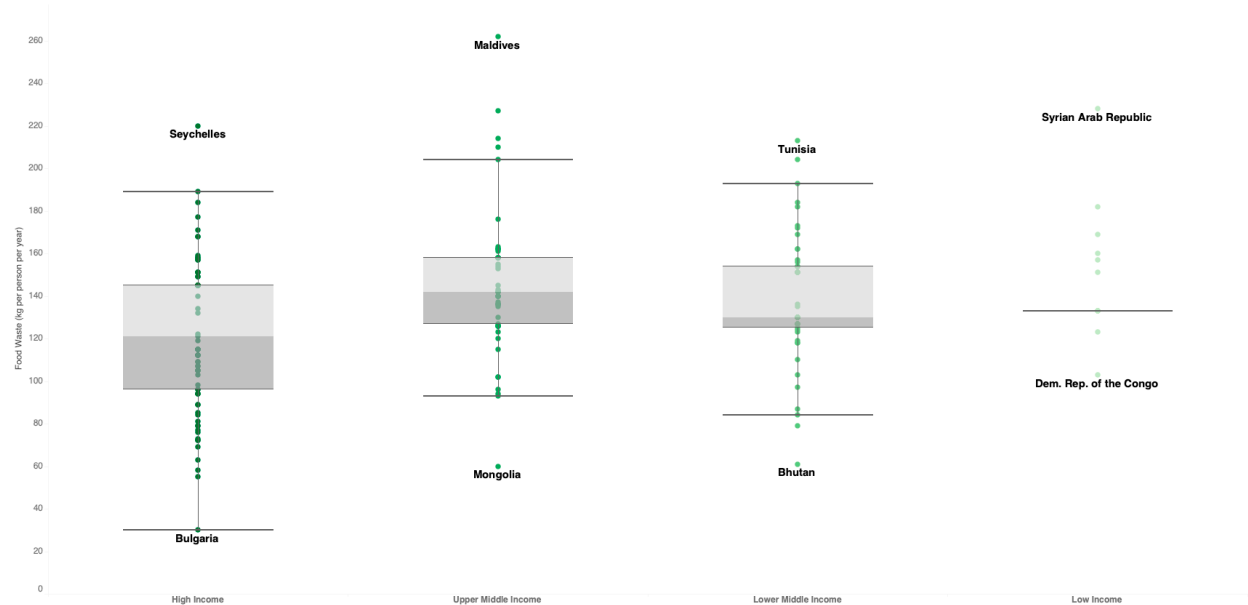


Figure 8. Food Waste Distribution by Income Group

10.7.1 Purpose

This chart evaluates **how food waste intensity (kg/person/year) varies across different income groups and assesses whether income level is a meaningful determinant of food waste behaviour**. The objective is to identify differences in central tendencies, variability, and outlier patterns among income categories.

10.7.2 Methodology

Countries were grouped using World Bank income classifications and visualized using **box plots to compare the distribution of household food waste per person across income groups**. Median values, interquartile ranges, overall spread, and notable outliers were examined to assess variability within and between groups.

10.7.3 Key Findings

Upper Middle Income countries recorded the highest median food waste intensity at approximately 142 kg/person/year, while the remaining income groups showed relatively similar

median levels. However, substantial variation and multiple outliers exist within each income group, indicating that **income level alone does not fully explain food waste behaviour and that other factors may play a significant role.**

10.7.4 Validation

The boxplot analysis includes 212 countries with valid income group classifications, excluding 21 countries where income group data was unavailable. **Distribution patterns, medians, quartiles, and outliers were reviewed to confirm consistency across income categories.**

10.7.5 Business / Sustainability Implication

The results suggest that food waste reduction strategies should not rely solely on economic classifications, as countries within the same income group can exhibit significantly different food waste outcomes. Effective interventions should **incorporate behavioural, cultural, policy, and food-system factors alongside income levels to achieve meaningful reductions in food waste.**

10.8 Chart 8: Temperature vs Food Waste per Capita

Warmer Countries Tend to Generate More Food Waste per Person

Temperature is moderately associated with food waste intensity ($r = 0.38$), although climate alone explains only a small portion of the variation ($R^2 = 0.14$)

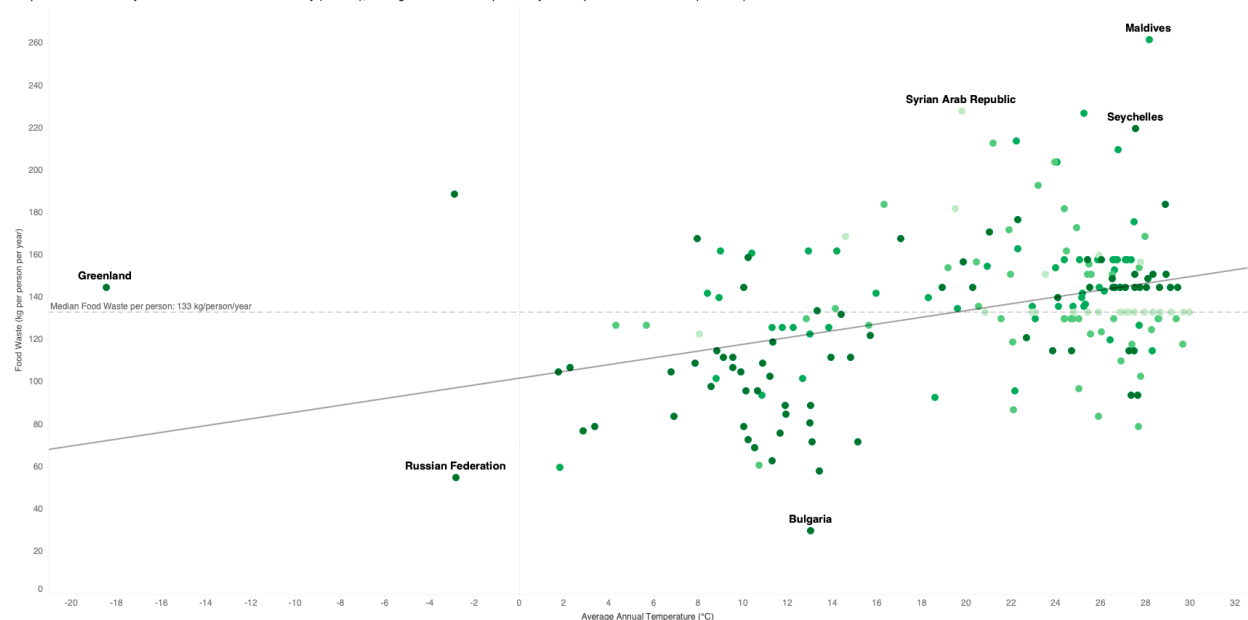


Figure 9. Temperature vs Food Waste per Capita

10.8.1 Purpose

This chart examines the **relationship between average annual temperature and household food waste intensity (kg/person/year) to determine whether climate conditions influence food waste behaviour.** The objective is to assess whether warmer countries tend to generate higher levels of food waste per person.

10.8.2 Methodology

A **scatter plot** was created using average annual temperature as the independent variable and household food waste per capita as the dependent variable. **Correlation analysis and a linear trend line were used to quantify the strength and direction of the relationship**, with notable outlier countries highlighted for interpretation.

10.8.3 Key Findings

The analysis shows a moderate positive relationship between temperature and food waste intensity ($r = 0.38$), indicating that **warmer countries generally tend to produce more food waste per person**. However, the explanatory power is relatively low ($R^2 \approx 0.14$), suggesting that **temperature accounts for only a small portion of the variation and that other factors also influence food waste behaviour**.

10.8.4 Validation

The relationship was validated using matched UNEP food waste and Trading Economics temperature data across countries with available records. Correlation coefficients, trend line patterns, and outlier observations such as Maldives, Seychelles, Greenland, and Bulgaria were reviewed to confirm the consistency of the findings.

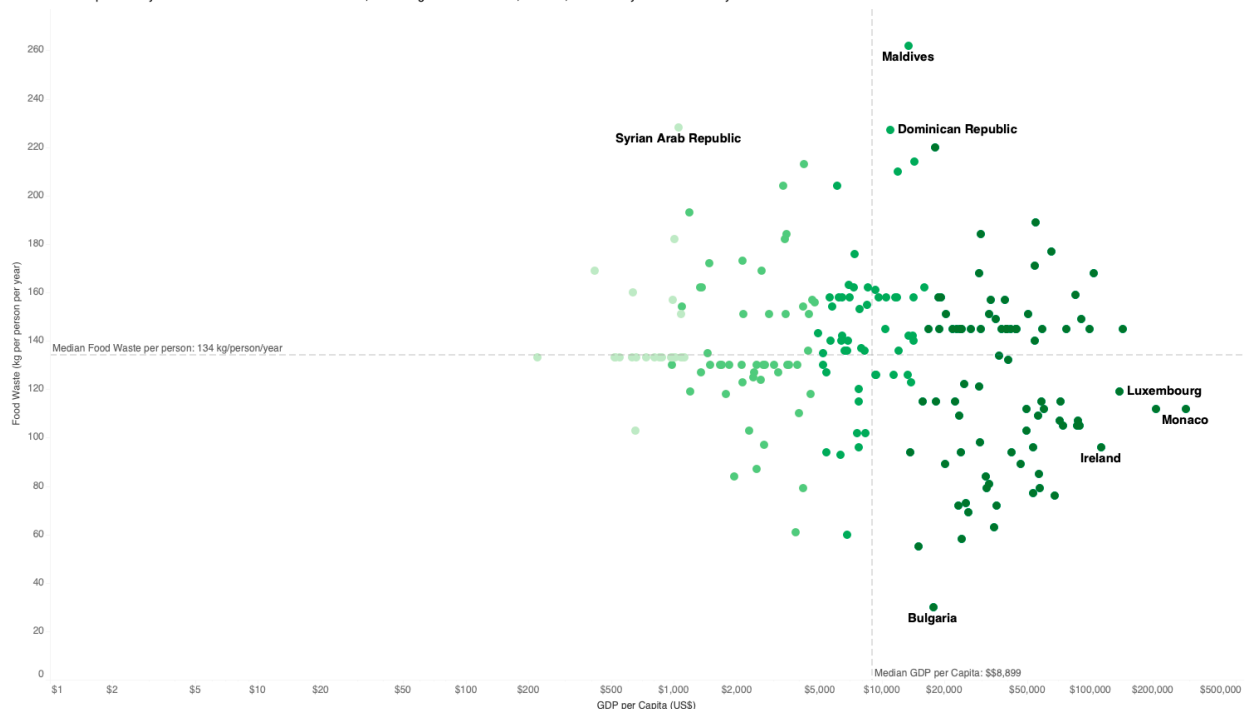
10.8.5 Business / Sustainability Implication

The results suggest that climate conditions may contribute to food waste patterns, potentially through food preservation challenges, spoilage rates, and consumption behaviour in warmer environments. However, because temperature alone explains limited variation, policymakers and sustainability practitioners should **combine climate considerations with behavioural, economic, and food-system interventions when designing food waste reduction strategies**.

10.9 Chart 9: GDP vs Food Waste per Capita

Economic Wealth Shows Little Relationship with Food Waste Intensity

Wealth alone explains very little variation in food waste outcomes, indicating that behavioural, climatic, and food-system factors may be more influential



10.9.1 Purpose

This chart examines whether **economic wealth, measured by GDP per capita, influences food waste intensity across countries**. The objective is to determine whether wealthier nations consistently generate more food waste per person than less affluent countries.

10.9.2 Methodology

A **scatter plot** was created using **GDP per capita (US\$)** on a **logarithmic scale** and household food waste per capita (kg/person/year) as the outcome variable. **Median reference lines were added to identify country clusters**, while correlation and coefficient of determination ($R^2 \approx 0.03$) were used to assess the strength of the relationship.

10.9.3 Key Findings

The analysis reveals **a very weak relationship between GDP per capita and food waste intensity**, with economic wealth explaining only about 3% of the variation in food waste outcomes. Significant differences exist among countries with similar income levels, **indicating that factors beyond wealth—such as consumer behaviour, food systems, culture, and climate—play a larger role in determining food waste levels.**

10.9.4 Validation

The analysis was conducted using the subset of 212 countries with valid income group classifications and the finding is supported by the **broad dispersion of countries across all GDP levels and the absence of a clear upward or downward trend**. Outliers such as Maldives, Syrian Arab Republic, Dominican Republic, Luxembourg, Monaco, and Ireland demonstrate that countries with comparable economic wealth can exhibit substantially different food waste outcomes.

10.9.5 Business / Sustainability Implication

The results suggest that increasing national wealth does not automatically lead to higher or lower food waste levels. Policymakers and sustainability leaders should therefore **prioritise behavioural change, food system efficiency, consumer education, and waste reduction programmes rather than relying solely on economic development as a driver of food waste reduction.**

10.10 Chart 10: Priority Intervention Countries Map

Geographic Distribution of Priority Intervention Countries

Countries exceeding both median food waste volume and median food waste per capita are identified as priority intervention hotspots.



Figure 11. Priority Intervention Countries Map

10.10.1 Purpose

This chart **identifies countries that represent the greatest opportunity for food waste reduction by highlighting those with both above-median total food waste volume and above-median food waste intensity (kg/person/year)**. The objective is to provide a geographic view of where interventions could potentially deliver the largest sustainability impact.

10.10.2 Methodology

Countries were classified as priority intervention hotspots if they **exceeded the global median for both total food waste tonnes and food waste per capita**. These countries were then visualized on a **choropleth map** to assess their **geographic distribution and identify regional patterns**.

10.10.3 Key Findings

Priority intervention countries are distributed across North America, South America, Africa, the Middle East, Asia, and Oceania, demonstrating that **food waste is a global challenge rather than a region-specific issue**. Major hotspots include **large-population countries such as China, the United States, Brazil, Indonesia, Pakistan, and Nigeria, where high waste volumes are combined with above-average waste intensity**.

10.10.4 Validation

Priority countries identified through the median-threshold approach were cross-checked against the Priority Intervention Risk Score rankings to ensure consistency between classification and prioritisation results. The classification was validated using the master dataset and median-based thresholds applied consistently across all countries with complete data. **Geographic results align with earlier analyses showing that population size is a strong driver of total food waste while behavioural and contextual factors contribute to higher per-capita waste levels**.

10.10.5 Business / Sustainability Implication

The map provides a practical framework for prioritising food waste reduction investments, policies, and awareness programmes in countries where both scale and intensity of waste are significant. **Focusing resources on these hotspots can maximise environmental benefits, reduce pressure on food systems, and accelerate progress toward global sustainability and food security goals**.

11. TABLEAU DASHBOARDS

11.1 Dashboard Design Approach

11.1.1 Audience

The dashboards were designed for policymakers, sustainability practitioners, business leaders, and data analysts who require both strategic insights and analytical transparency. Visualizations, KPIs, and interactive features were structured to support quick decision-making for stakeholders while providing sufficient detail for technical audiences to validate and explore the findings.

11.1.2 Business Questions

The dashboard suite was designed to answer five key questions:

- How much food waste is generated globally?
- What sectors contribute most to food waste?
- What factors are associated with food waste levels?

- Which countries represent the highest intervention priority?
- Where are food waste hotspots geographically concentrated?

These questions support evidence-based decision-making by identifying areas where food waste reduction initiatives can deliver the greatest environmental and social impact.

11.2 Dashboard 1: Global Food Waste Overview

GLOBAL FOOD WASTE OVERVIEW

Scale, Top Contributors, and Sector composition of global food waste

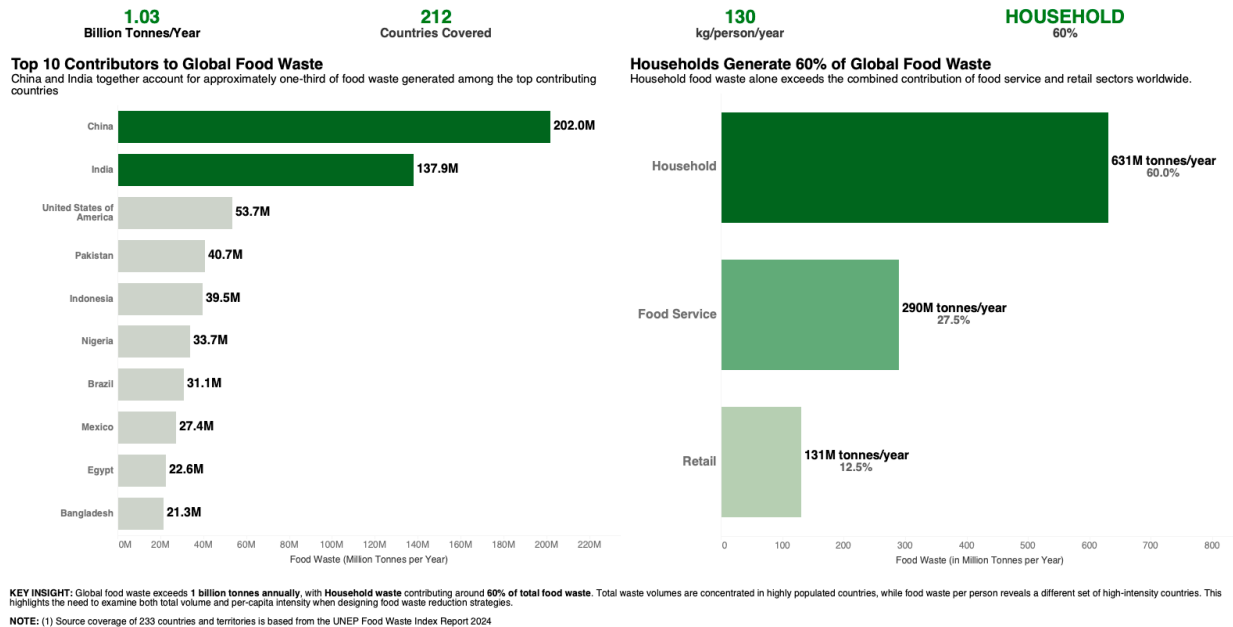


Figure 12. Global Food Waste Overview

11.2.1 Components

Dashboard 1 comprises the following charts into a single executive view:

- **KPI Cards** highlighting total global food waste, countries covered, average food waste per person, and the largest waste sector
- **Chart 1 – Global Food Waste Scale and Coverage**
- **Chart 2 – Top 10 Contributors to Global Food Waste (Horizontal Bar Chart)**
- **Chart 5 – Food Waste by Sector Composition (Bar Chart)**

Together, these visualizations provide a high-level understanding of the scale of global food waste, the countries contributing the largest waste volumes, and the relative contribution of the household, food service, and retail sectors.

11.2.2 Key Findings

The analysis shows that global food waste exceeds **1 billion tonnes per year** across **233 countries**, with **households** accounting for approximately **60%** of total food waste, exceeding the combined contribution of the food service and retail sectors.

Food waste volumes are highly concentrated in populous countries, with **China and India contributing approximately one-third of total food waste among the top contributing countries**. This highlights the importance of targeting high-volume countries while also considering per-capita waste patterns when designing intervention strategies.

11.3 Dashboard 2: Major Drivers of Food Waste

MAJOR DRIVERS OF FOOD WASTE

Exploring the strongest demographic and climate-related factors associated with food waste outcomes

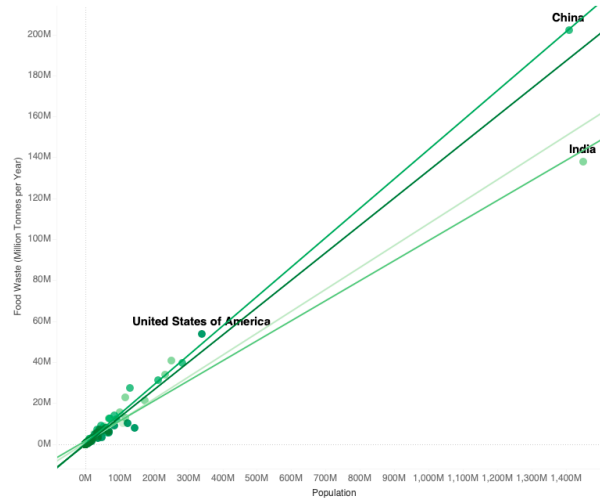
Select Income Group Select Country

POPULATION
 $R^2 \approx 0.89$

Strongest Driver of Total Food Waste

Population Size Is a Strong Predictor of Total Food Waste Generation

Population size is strongly associated with total food waste generation across countries ($R^2 \approx 0.89$).

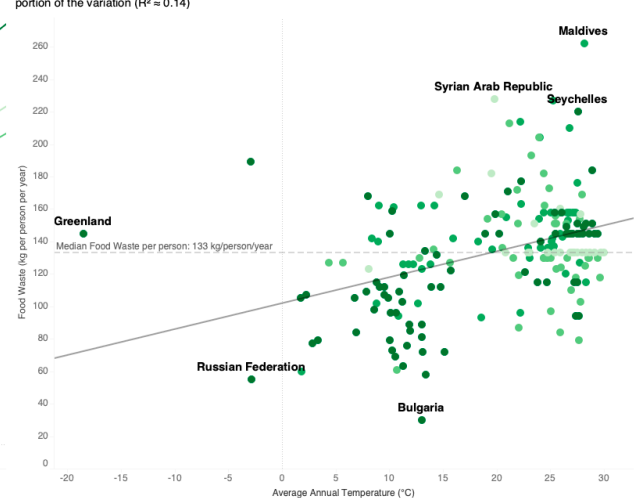


TEMPERATURE
 $r = 0.38$

Strongest Driver of Food Waste per Person

Warmer Countries Tend to Generate More Food Waste per Person

Temperature is moderately associated with food waste intensity ($r = 0.38$), although climate alone explains only a small portion of the variation ($R^2 \approx 0.14$).



KEY INSIGHT: Population Size is the strongest predictor of total food waste volume, while Average Temperature exhibits the strongest association with food waste intensity per person. Together, these findings suggest that demographic scale and climate conditions influence food waste outcomes more strongly than economic factors.

NOTES: (1) Driver analysis uses countries with available matched World Bank, temperature, and UNEP data. Coverage varies for each variable. (2) Income-group analyses exclude 21 countries with unavailable World Bank income classifications (n = 212 countries).

Figure 13. Major Drivers of Food Waste

11.3.1 Components

Dashboard 2 comprises the following charts in a single analytical view:

- **KPI Cards** highlighting the strongest demographic and environmental drivers of food waste and the strength of their relationships with food waste outcomes
- **Chart 4A – Population vs Total Food Waste (Scatter Plot)**
- **Chart 8 – Temperature vs Food Waste per Capita (Scatter Plot)**

Together, these visualizations examine the **relationship between food waste and selected demographic and environmental factors**, helping identify which variables are most strongly associated with food waste volume and food waste intensity across countries.

Interactive filters for Income Group and Country enable users to explore patterns across different country segments and individual countries.

11.3.2 Key Findings

The analysis indicates that **population size is the strongest predictor of total food waste generation**, exhibiting a very strong positive relationship with food waste volume ($R^2 \approx 0.89$). In contrast, average annual temperature demonstrates the strongest association with food waste per person ($r = 0.38$), suggesting that **warmer countries tend to generate higher levels of household food waste per capita**, although climate alone explains only a modest portion of the variation.

These findings highlight that both demographic scale and environmental conditions influence food waste outcomes, but population remains the dominant driver of total food waste generation globally.

11.4 Dashboard 3: Income & Economic Factors

ECONOMIC WEALTH SHOWS LITTLE RELATIONSHIP WITH FOOD WASTE INTENSITY

Examining whether wealth and income levels explain differences in food waste behaviour

Select Income Group Select Country

(Multiple values)

Highlight Country Legend

INCOME GROUP
121–142 kg/person/year
Similar Median Food Waste across Groups

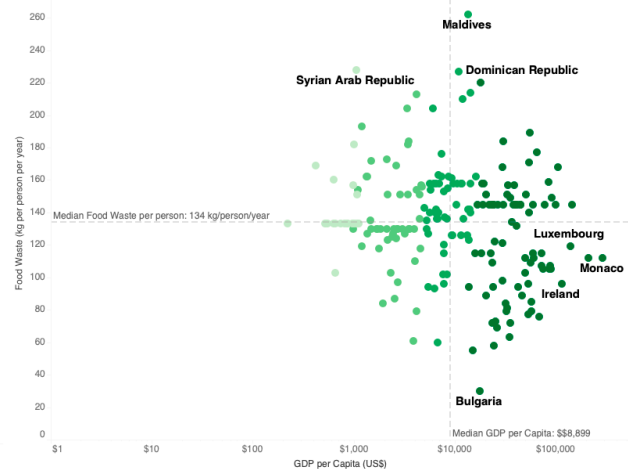
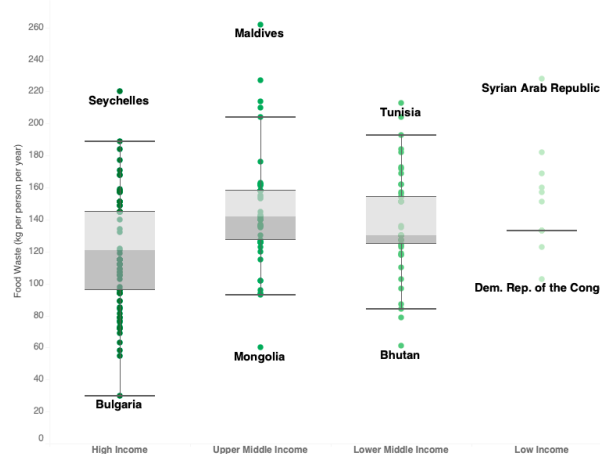
GDP PER CAPITA
 $R^2 \approx 0.03$
Very Weak Relationship

Food Waste Varies Significantly Within Each Income Group

Upper-middle-income countries record the highest median food waste intensity (142 kg/person/year), while wide distributions and outliers across all income groups suggest that income alone does not fully explain food waste behaviour.

Economic Wealth Shows Little Relationship with Food Waste Intensity

Wealth alone explains very little variation in food waste outcomes, indicating that behavioural, climatic, and food-system factors may be more influential



KEY INSIGHT: Income level and GDP per Capita explain relatively little variation in food waste outcomes.

Countries within the same income group often display substantial differences in food waste intensity, suggesting that behavioural, cultural, and food-system factors may be more influential than economic wealth alone.

NOTES: (1) Driver analysis uses countries with available matched World Bank, temperature, and UNEP data. Coverage varies for each variable. (2) Income-group analyses exclude 21 countries with unavailable World Bank income classifications (n = 212 countries).

Figure 14. Income & Economic Factors

11.4.1 Components

Dashboard 3 comprises the following charts in a single analytical view:

- **KPI Cards** highlighting median food waste intensity by income group and the relationship between GDP per capita and food waste intensity
- **Chart 7 – Food Waste Distribution by Income Group (Box Plot)**
- **Chart 9 – GDP vs Food Waste per Capita (Scatter Plot)**

Together, these visualizations **examine whether income level and economic wealth influence food waste intensity across countries.**

Interactive filters for Income Group and Country allow users to compare food waste patterns within and across economic segments.

11.4.2 Key Findings

The analysis indicates that **income level and GDP per capita explain relatively little variation in food waste intensity**, with GDP per capita exhibiting a very weak relationship with household food waste per person ($R^2 \approx 0.03$).

While **upper-middle-income countries record the highest median food waste intensity**, **substantial variation exists within each income group**, suggesting that behavioral, cultural, climatic, and food-system factors may play a more significant role than economic wealth alone in influencing food waste outcomes.

11.5 Dashboard 4: Food Waste Risk & Opportunity Profile

FOOD WASTE RISK & OPPORTUNITY PROFILE

Identifying countries with the greatest potential impact for food waste reduction and sustainability interventions

Highest Total Food Waste Country

CHINA
202M tonnes/year

Highest Food Waste Intensity

MALDIVES
262 kg/person/year

Largest Waste Sector

HOUSEHOLD
60%

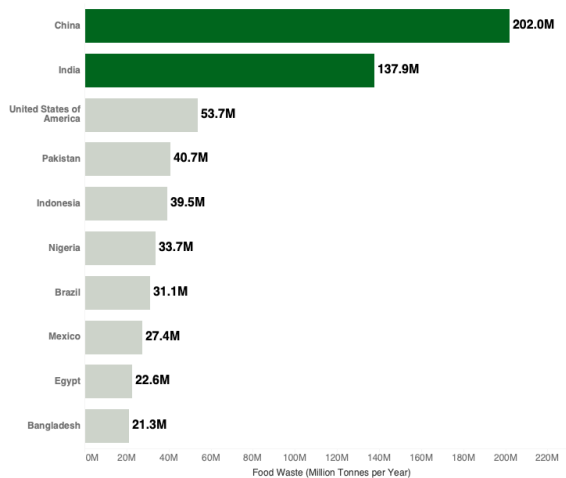
High Risk Countries

60

Above Median Waste Volume & Waste Intensity

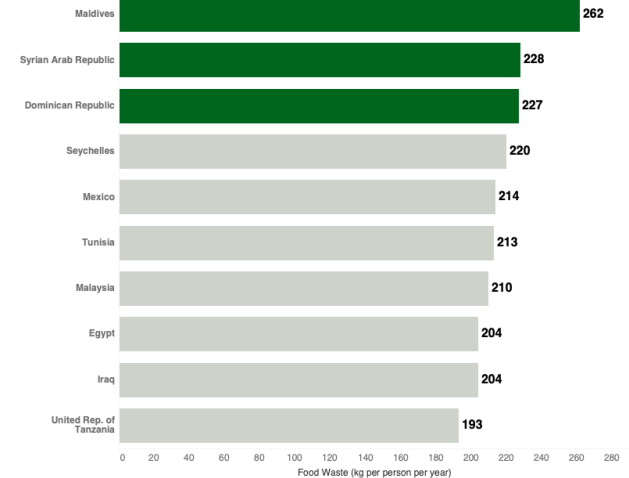
Top 10 Contributors to Global Food Waste

China and India together account for approximately one-third of food waste generated among the top contributing countries



Countries Generating the Highest Food Waste per Person

Countries with the highest food waste intensity are not necessarily the most populous or wealthiest nations



KEY INSIGHT: Countries with large populations contribute disproportionately to global food waste volumes, while the highest food waste intensity countries are not necessarily the wealthiest or most populous. These findings suggest that behavioural, cultural, and food-system factors may play a significant role in shaping food waste outcomes.

NOTE: Driver analysis uses countries with available matched World Bank, temperature, and UNEP data. Coverage varies for each variable.

Figure 14. Food Waste Risk & Opportunity Profile

11.5.1 Components

Dashboard 4 comprises the following charts in a single analytical view:

- **KPI Cards** highlighting the highest total food waste country, highest food waste intensity country, largest waste sector, and the number of high-risk countries
- **Chart 1 – Top 10 Contributors to Global Food Waste (Horizontal Bar Chart)**
- **Chart 2 – Countries Generating the Highest Food Waste per Person (Horizontal Bar Chart)**

Together, these visualizations compare food waste from both a **volume perspective** and an **intensity perspective**, enabling the identification of countries that present the greatest opportunities for targeted food waste reduction and sustainability interventions.

11.5.2 Key Findings

The analysis shows that countries with the largest total food waste volumes are primarily highly populated nations, with **China generating approximately 202 million tonnes of food waste annually**, followed by India and the United States.

In contrast, the countries with the highest food waste intensity per person, led by **Maldives at 262 kg/person/year**, are not necessarily the most populous or wealthiest, indicating that food waste outcomes are influenced by factors beyond population size and economic development.

The dashboard also identified **60 high-risk countries** exhibiting both above-median food waste volume and food waste intensity, highlighting **priority areas for policy intervention and food waste reduction initiatives**.

11.6 Dashboard 5: Food Waste Intervention Hotspots

PRIORITY COUNTRIES FOR FOOD WASTE REDUCTION

60 Priority Countries **212** Countries Covered Global Food Waste **1.05** Billion Tonnes/Year Largest Waste Sector **HOUSEHOLD** 60%

Geographic Distribution of Priority Intervention Countries

Countries exceeding both median food waste volume and median food waste per capita are identified as priority intervention hotspots.



KEY INSIGHT: Priority intervention countries are distributed across North America, South America, Africa, the Middle East, Asia, and Oceania, demonstrating that food waste is a global challenge rather than a regional issue. Countries exceeding both median food waste volume and median food waste per capita represent the greatest opportunity for targeted waste reduction initiatives, where interventions can achieve both large-scale impact and meaningful behavioural change.

NOTE: Priority countries are defined as countries exceeding both the global median food waste volume and global median food waste per capita.

Figure 15. Food Waste Intervention Hotspots

11.6.1 Components

Dashboard 5 comprises the following visualizations in a single geographical view:

- **KPI Cards** highlighting the number of priority countries, countries covered, total global food waste, and the largest waste sector
- **Chart 10 – Priority Intervention Countries Map (Choropleth Map)**

Together, these visualizations identify countries that exceed both the global median food waste volume and median food waste intensity, enabling policymakers and sustainability practitioners to prioritize locations where food waste reduction efforts may achieve the greatest impact.

11.6.2 Key Findings

The analysis identified **60 priority intervention countries** that exhibit both above-median food waste volume and above-median food waste intensity, representing locations with the greatest potential for meaningful food waste reduction outcomes.

These countries are distributed across **North America, South America, Africa, the Middle East, Asia, and Oceania**, demonstrating that **food waste is a global challenge rather than a region-specific issue**. By targeting countries with both high waste volumes and high waste intensity, interventions can potentially deliver significant environmental, economic, and social benefits while supporting progress toward **UN SDG 12.3**.

12. STATISTICAL ANALYSIS

12.1 Descriptive Statistics

Descriptive statistical measures, including the mean, median, quartiles, and standard deviation, were used during the exploratory analysis phase to understand the distribution, variability, and central tendencies of food waste and related indicators across countries.

These statistics informed the interpretation of box plots, the identification of outliers, the development of risk thresholds, and the selection of variables for subsequent correlation analysis and predictive modeling.

12.2 Correlation Analysis

12.2.1 Correlation Analysis Summary

Given the primary target variable, **Total Food Waste (kg per Capita per Year)**, the Pearson correlation was computed for all variables, while Spearman correlation was explicitly calculated and documented for **GDP per Capita**, **Average Temperature**, and **Tourists per Resident**.

Variable	Pearson Correlation Analysis	Spearman Correlation Analysis	Analysis / Conclusion
Average Temperature	0.380	0.328	Moderate positive relationship and the strongest correlation observed among the tested variables. Countries with higher average temperatures tend to have higher food waste per capita, although temperature alone explains only a portion of the variation.
GDP per Capita	-0.158	-0.142	Weak negative relationship. Economic wealth shows a statistically significant but limited association with food waste per capita, suggesting GDP alone is not a strong predictor.
Population Density	-0.021	Not calculated	Negligible relationship. Population density appears to have little influence on food waste per capita at the country level.
Tourism Arrivals	-0.127	Not calculated	Weak negative relationship. Absolute tourism activity does not appear to be a meaningful driver of food waste per capita across countries.
Tourists per Resident (Tourism Intensity)	-0.035	-0.006	Virtually no relationship. Countries with higher tourism intensity do not necessarily generate more food waste per capita.
Access to Electricity	-0.002	Not calculated	No meaningful relationship. Electricity access appears unrelated to food waste per capita within the available dataset.

12.2.2 Overall Findings

The correlation analysis indicates that **Average Temperature** demonstrated the strongest association with food waste per capita, although the relationship remains moderate rather than strong. All other variables exhibited weak or negligible correlations, suggesting that food waste is influenced by a combination of factors rather than a single dominant driver. These findings

supported the conclusion reached during the predictive modeling phase that the available country-level variables have limited linear explanatory power and that additional behavioral, cultural, policy, or food system variables may be required to improve predictive performance.

12.3 Priority Intervention Framework

To identify countries where food waste reduction initiatives could deliver the greatest impact, a Priority Intervention Framework was developed. Countries were evaluated using both total food waste volume (tonnes/year) and food waste intensity (kg/person/year), ensuring that intervention priorities reflected both scale and behavioural significance.

12.3.1 Intervention Criteria

Countries exceeding the global median for both total food waste volume and food waste intensity were classified as Priority Intervention Countries. This median-based approach provides a robust, distribution-independent method for identifying countries that perform above the global norm on both measures.

12.3.2 Risk Score Calculation

A **Priority Intervention Risk Score** was developed to **rank countries according to their relative intervention opportunity**. The score was calculated as the product of each country's food waste volume ratio and food waste intensity ratio relative to their respective global median values:

$$\text{Risk Score} = \left(\frac{\text{Total Food Waste}}{\text{Median Total Food Waste}} \right) \times \left(\frac{\text{Food Waste Per Capita}}{\text{Median Food Waste Per Capita}} \right)$$

Higher scores indicate countries where both the scale and intensity of food waste are elevated, suggesting greater potential impact from targeted food waste reduction interventions. The score was used as a prioritisation and ranking mechanism rather than a predictive risk model.

12.4 Feature Assessment

A feature assessment was conducted to evaluate the relevance and predictive potential of the socioeconomic, demographic, environmental, infrastructure, and tourism indicators included in the master dataset. This assessment combined correlation analysis, exploratory data analysis, data quality considerations, and predictive model testing to determine which variables contributed meaningful information for explaining variations in household food waste.

The results indicated that the following variables demonstrated relatively stronger associations with household food waste and were selected for further evaluation during predictive modeling:

- Average Temperature
- GDP per Capita
- Food Service Waste (kg per capita)
- Population Density

Variables such as Tourism Intensity and Access to Electricity exhibited weak relationships with household food waste and showed limited predictive value.

To improve model performance and address skewed data distributions, GDP per Capita and Population Density were subsequently transformed using a logarithmic scale during the predictive modeling phase. These findings informed feature selection, transformation, and refinement while highlighting the complex and multi-factor nature of food waste across countries.

13. PREDICTIVE MODELING

13.1 Business Objective

The objective of the predictive modeling phase was to develop a model to estimate the **household food waste per person (kg/person)** using selected socioeconomic, demographic, and environmental indicators. The analysis sought to identify key drivers of household food waste and support more targeted food waste reduction strategies.

13.2 Modeling Approach

The dataset was partitioned into **training and testing subsets using an 80:20 split with a fixed random state of 27**, ensuring consistent and reproducible model evaluation. A Baseline Model was first established to provide a performance benchmark, and the same training and testing datasets were used across the **Baseline, Linear Regression, and Random Forest** models to enable fair comparisons. Feature selection was guided by exploratory analysis, correlation results, and business relevance.

To improve model effectiveness, logarithmic transformations were applied to highly skewed variables to reduce the influence of extreme values and improve model stability:

- GDP per Capita
- Total Population
- Population Density
- International Tourism Arrivals

while income groups were encoded into numerical values to create a machine-learning-ready dataset.

13.3 Model Performance Metrics

13.3.1 MAE (Mean Absolute Error)

It measures the average absolute difference between the actual value and the predicted value. There is no universal "good" or "bad" MAE because it depends on the scale of the target variable.

Value	Interpretation
0	Perfect prediction
> 0	Some prediction error
Lower is better	Always

13.3.2 RMSE (Root Mean Squared Error)

It measures the average prediction error, but larger errors are penalized more heavily because errors are squared first.

Value	Interpretation
0	Perfect prediction
Lower is better	Always

13.3.3 R² (R-Squared or Coefficient of Determination)

It measures the percentage of variation in household food waste that the model can explain.

R ² Value	Interpretation
1.00	Perfect prediction

R ² Value	Interpretation
0.80–1.00	Excellent
0.60–0.80	Strong
0.40–0.60	Good
0.20–0.40	Reasonable
0.00–0.20	Weak
< 0.00	Worse than predicting the mean

13.4 Baseline Model

13.4.1 Features Used

The Baseline Model did not use any predictor variables. A DummyRegressor with the mean strategy was implemented, which predicts the average household food waste value for all countries regardless of their characteristics.

13.4.2 Results

The Baseline Model achieved the following results:

Data Model	MAE	RMSE	R ²
Baseline	15.22	26.58	-0.014

13.4.3 Interpretation

The Baseline Model established the minimum performance benchmark for subsequent predictive models. The negative R² indicates that predicting the global average household food waste explained virtually none of the variation across countries, highlighting the need for models that incorporate country-level characteristics.

Metric	Result	Interpretation
MAE	15.22 kg/capita/year	On average, predictions are off by about 15 kg per person per year
RMSE	26.58 kg/capita/year	Larger errors exist for some countries
R ²	-0.014	Slightly worse than simply using the overall mean

13.4.4 What would success look like?

Data Model	MAE	RMSE	R ²
Baseline Model	MAE = 15.22	< 20	-0.014
Linear Regression	MAE < 15.22	< 20	> 0.10
Random Forest	MAE significantly lower than 15.22	< 20	> 0.10

13.5 Linear Regression

13.5.1 Features Used

Three Linear Regression iterations were evaluated.

Round 1 used eight numerical features (food service waste, retail waste, average temperature, GDP per capita, population density, tourism arrivals, access to electricity, and total population) together with income group and region.

Following correlation analysis, logarithmic transformation of skewed variables, and feature selection, Round 2 retained four numerical features (average temperature, log GDP per capita, log population density, and food service waste per capita) and one categorical feature (income group).

Round 3 tested the addition of household region to assess whether geographic differences would improve model performance.

13.5.2 Results

Data Model	Round	# of numerical features	# of categorical features	MAE	RMSE	R ²
Baseline	-	-	-	15.22	26.58	-0.014
Linear Regression	1	8	2	18.22	31.23	-0.401
Linear Regression	2	4	1	16.57	26.34	0.004
Linear Regression	3	4	2	17.04	28.45	-0.162

13.5.3 Interpretation

The initial model demonstrated that the full set of predictors did not generalize well to unseen countries.

Feature engineering and feature selection improved performance by reducing noise and retaining the most relevant variables, resulting in Round 2 achieving the **best overall performance among the Linear Regression models**, with the lowest RMSE and the only positive R² value. Although Round 2 did not outperform the Baseline Model in MAE, it produced more consistent predictions and explained a small portion of the variation in household food waste.

The inclusion of household region in Round 3 reduced model performance, suggesting that regional differences observed during exploratory analysis did not translate into stronger predictive power.

Overall, the limited R² values indicate that household food waste is influenced by additional behavioural, cultural, and policy-related factors not captured in the available dataset.

13.6 Random Forest Regression

13.6.1 Features Used

The Random Forest Regression model was developed using the same refined feature set identified through the Linear Regression feature selection process.

The predictors included four numerical features: **average temperature**, **log-transformed GDP per capita**, **log-transformed population density**, and **food service waste per capita**, together with one categorical feature: **income group**. This feature set was selected based on correlation analysis, feature engineering, and model refinement performed during the Linear Regression phase.

13.6.2 Results

Data Model	Round	n_estimators	max_depth	min_samples	MAE	RMSE	R ²
Baseline	-	-	-	-	15.22	26.58	-0.014
Random Forest	1	300	5	5	15.08	26.14	0.019
Random Forest	2	500	8	3	15.90	26.80	-0.031

13.6.3 Interpretation

Random Forest was evaluated to determine whether non-linear relationships between country-level indicators and household food waste could be captured more effectively than Linear Regression.

Round 1 delivered the best overall Random Forest performance, achieving lower MAE and RMSE values than both the Baseline Model and the final Linear Regression model, while also producing the highest R^2 value among all models tested.

Increasing model complexity in Round 2 did not improve performance and resulted in weaker evaluation metrics, indicating potential overfitting and reduced generalization to unseen countries. Consequently, the Round 1 Random Forest model was selected as the final predictive model for this study.

Feature importance analysis further identified **food service waste per capita**, **population density**, **GDP per capita**, and **average temperature** as the most influential predictors of household food waste.

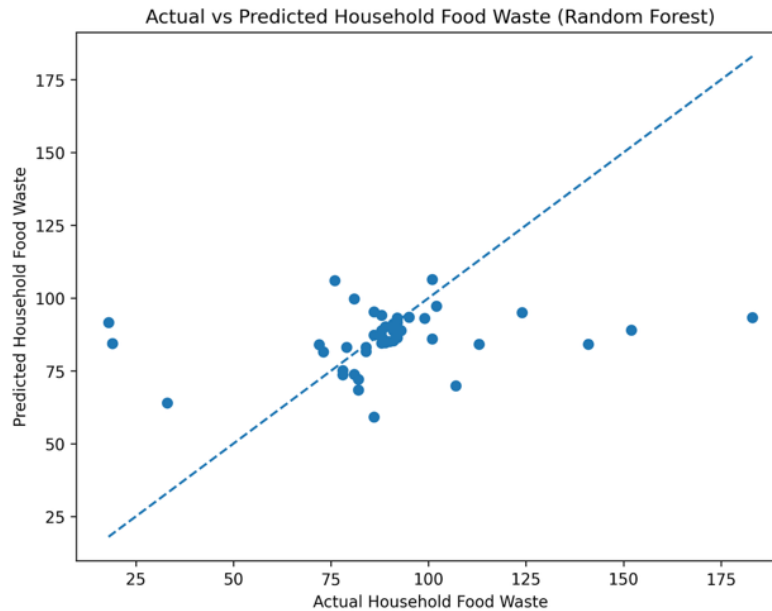
13.7 Model Comparison

Several predictive models were evaluated, including a Baseline Model, multiple Linear Regression iterations, and Random Forest Regression. The Linear Regression models underwent feature engineering, logarithmic transformation, feature selection, and region testing. While these refinements improved model performance, the best Linear Regression model achieved only marginal improvements over the Baseline Model, indicating limited linear relationships between the available predictors and household food waste.

Random Forest Regression (Round 1) was selected as the final predictive model because it achieved the best overall performance, with the lowest MAE (15.08), lowest RMSE (26.14), and highest R^2 (0.019) among all models evaluated. The model was able to capture non-linear relationships between the predictors and household food waste that were not effectively identified by Linear Regression.

Data Model	Round	MAE	RMSE	R^2	Result
Baseline		15.22	26.58	-0.014	
Linear Regression	1	18.22	31.23	-0.401	
Linear Regression	2	16.57	26.34	0.004	
Linear Regression	3	17.04	28.45	-0.162	
Random Forest	1	15.08	26.14	0.019	Best Overall Performing Model
Random Forest	2	15.90	26.80	-0.031	

13.8 Best Performing Model



13.8.1 Strengths

The model achieved the best overall predictive performance and reduced prediction errors relative to both the Baseline and Linear Regression models. The Actual vs Predicted plot shows that **predictions generally followed the overall trend of the observed values, particularly for countries with average household food waste levels**. The model also provided feature importance rankings, offering additional insight into the factors most strongly associated with household food waste.

13.8.2 Weaknesses

Despite outperforming the other models, the model's explanatory power remained limited, as reflected by the low R^2 value. The Actual vs Predicted plot indicates that **the model tended to predict values closer to the global average and was less effective at accurately predicting countries with exceptionally high or low household food waste**. This suggests that important behavioural, cultural, policy, and consumption-related factors influencing food waste were not captured in the available dataset.

13.9 Prediction Examples

Country	Actual	Predicted	Prediction_Error	Absolute_Error
Guadeloupe	95	93.4929261	1.5070739	1.5070739
Greenland	79	83.1132901	-4.1132901	4.11329007
China, Hong Kong SAR	101	106.431393	-5.4313932	5.43139319
Paraguay	91	85.3332324	5.66676756	5.66676756
Niger	92	92.7603273	-0.7603273	0.76032733
Portugal	124	95.0095512	28.9904488	28.9904488
Italy	107	69.9367915	37.0632085	37.0632085
Montenegro	86	87.2951714	-1.2951714	1.29517143
Seychelles	183	93.3517647	89.6482353	89.6482353
Ghana	84	81.6632282	2.33677185	2.33677185

14. KEY FINDINGS AND INSIGHTS

14.1 Households Account for the Largest Share of Global Food Waste

Households contribute approximately **60% of total global food waste**, making them the largest source of food waste across all sectors. This highlights the importance of consumer-focused interventions, education, and behavioral change initiatives in achieving meaningful food waste reduction.

14.2 Food Waste Patterns Differ Across Income Groups

Food waste intensity varies across income groups, with **Upper-Middle-Income countries exhibiting the highest median household food waste per person**. However, the wide variation within income groups suggests that factors beyond income, such as consumption habits, culture, and food systems, also influence food waste outcomes.

14.3 Population Is Strongly Associated with Total Food Waste Generation

Population size showed the strongest relationship with total food waste generation ($R^2 \approx 0.89$), indicating that countries with larger populations tend to generate substantially higher food waste volumes. This emphasizes the need to consider population scale when prioritizing food waste reduction efforts and resource allocation.

14.4 GDP Is More Closely Associated with Food Waste Intensity Than Temperature

GDP per Capita demonstrated a stronger relationship with household food waste intensity than temperature, although both relationships were relatively weak. This suggests that economic development may influence food waste behavior, but other factors are likely more important drivers of per-capita food waste.

14.5 High-Risk Countries Present Significant Intervention Opportunities

The analysis identified **60 priority intervention countries** with both above-median food waste volume and food waste intensity, indicating significant opportunities for targeted action. Focusing resources on these countries may deliver the greatest impact in reducing food waste and advancing sustainability goals.

15. RECOMMENDATIONS

Based on the findings of this study, targeted recommendations are proposed for key stakeholders to strengthen food waste measurement, prioritize high-impact interventions, advance sustainability initiatives, and support evidence-based decision-making toward achieving **UN Sustainable Development Goal (SDG) 12.3**.

Audience	Actionable Measure 1	Actionable Measure 2
United Nations Environment Programme (UNEP)	Strengthen global food waste measurement frameworks by promoting standardized data collection and reporting methodologies across countries.	Prioritize support for high-risk countries identified in this study to accelerate progress toward SDG 12.3 through targeted food waste reduction programs.

Audience	Actionable Measure 1	Actionable Measure 2
Policymakers and Government Agencies	Develop household-focused food waste reduction campaigns, as households account for the largest share of global food waste.	Use data-driven targeting to prioritize interventions in countries and population segments exhibiting both high food waste volumes and high per-capita waste levels.
Sustainability Practitioners	Design behavior-change initiatives that encourage responsible food purchasing, meal planning, storage, and consumption practices at the household level.	Establish measurable food waste reduction targets and monitoring frameworks to track program effectiveness and sustainability outcomes.
Researchers	Expand future studies by incorporating additional variables such as food prices, urbanization, food insecurity, and waste management infrastructure to improve explanatory power.	Explore advanced machine learning and longitudinal approaches to better understand causal relationships and predict food waste trends over time.
Non-Governmental Organizations (NGOs)	Focus education and community outreach efforts on reducing household food waste through practical and culturally relevant interventions.	Partner with governments, businesses, and local communities to scale food waste awareness, redistribution, and food recovery initiatives in priority countries.
Students and Data Analysts	Apply data analytics and visualization techniques to sustainability challenges to support evidence-based decision-making and public awareness.	Contribute to open data, reproducible research, and innovative analytical approaches that improve understanding of global food waste and sustainability issues.

16. LIMITATIONS

16.1 Estimated Food Waste Data

The UNEP Food Waste Index contains modeled and estimated values for many countries rather than directly measured observations. As a result, the reported food waste figures may not fully reflect actual waste levels and should be interpreted as informed estimates.

16.2 Reporting Year Differences

Although data was standardized to the 2015–2024 period, the source datasets were not all reported in the same year. These temporal differences may affect comparability and introduce minor inconsistencies when analyzing relationships between variables.

16.3 Country Coverage Differences

Not all datasets covered the same set of countries, requiring country matching, exclusions, and data integration decisions during the development of the master dataset. Consequently, some countries and variables could not be included in the final analysis.

16.4 Missing Variables

The analysis focused on selected socioeconomic, demographic, environmental, tourism, and infrastructure indicators, but did not include potentially important factors such as food prices, urbanization, consumer behavior, waste management systems, and food insecurity. The omission of these variables may limit the ability to fully explain differences in food waste across countries.

16.5 Cross-Sectional Nature of Data

The study analyzed country-level data at a single point in time rather than examining changes over multiple years. As a result, the findings identify associations between variables but do not establish causation or reveal long-term trends in food waste behavior.

16.6 Predictive Model Constraints

The predictive models were developed using a relatively small country-level dataset and a limited set of available features. While the models provide useful directional insights, their predictive performance and generalizability may improve with larger datasets and additional explanatory variables.

17. FUTURE ENHANCEMENTS AND NEXT STEPS

17.1 Additional Datasets

Future research could incorporate additional variables such as food prices, urbanization rates, food insecurity indicators, waste management infrastructure, consumer behavior metrics, and government sustainability policies. These factors may provide a more comprehensive understanding of the drivers of food waste and improve explanatory power.

17.2 Time-Series Analysis

Expanding the analysis to include multiple years of food waste data would enable the study of trends, seasonality, and changes in food waste patterns over time. This would provide deeper insights into the effectiveness of policy interventions and progress toward **UN SDG 12.3**.

17.3 Advanced Predictive Modeling

Future studies could evaluate more advanced machine learning techniques, such as Gradient Boosting, XGBoost, Random Forest optimization, or ensemble models, to improve predictive performance and uncover more complex relationships between variables.

17.4 Geographic and Regional Analysis

Additional analysis at regional, sub-national, or city levels could help identify localized food waste patterns and support more targeted interventions. This would be particularly valuable for countries with significant population and socioeconomic diversity.

17.5 Enhanced Dashboard Capabilities

Future dashboard enhancements could include trend analysis, scenario modeling, country benchmarking, predictive forecasting, and progress tracking against food waste reduction targets. These features would provide stakeholders with more actionable insights for monitoring and decision-making.

18. CONCLUSION

This study successfully achieved its objectives by developing a consolidated global food waste dataset, identifying key drivers and patterns of food waste, creating interactive Tableau dashboards, and building predictive models to estimate household food waste. Through comprehensive data loading, cleaning, validation, and country matching activities, a high-quality and analysis-ready dataset was established, providing a reliable foundation for exploratory analysis and modeling.

The findings revealed that **households** account for the **largest share of global food waste**, **population is the strongest driver** of total food waste generation, and a subset of countries presents significant opportunities for targeted intervention.

While the predictive models demonstrated the potential to estimate household food waste using socioeconomic, demographic, and environmental indicators, the results also highlighted the importance of additional variables to further improve predictive performance.

Based on these insights, the study recommends strengthening food waste measurement, prioritizing household-focused interventions, and adopting data-driven strategies to support progress toward **UN Sustainable Development Goal 12.3** and broader sustainability objectives.